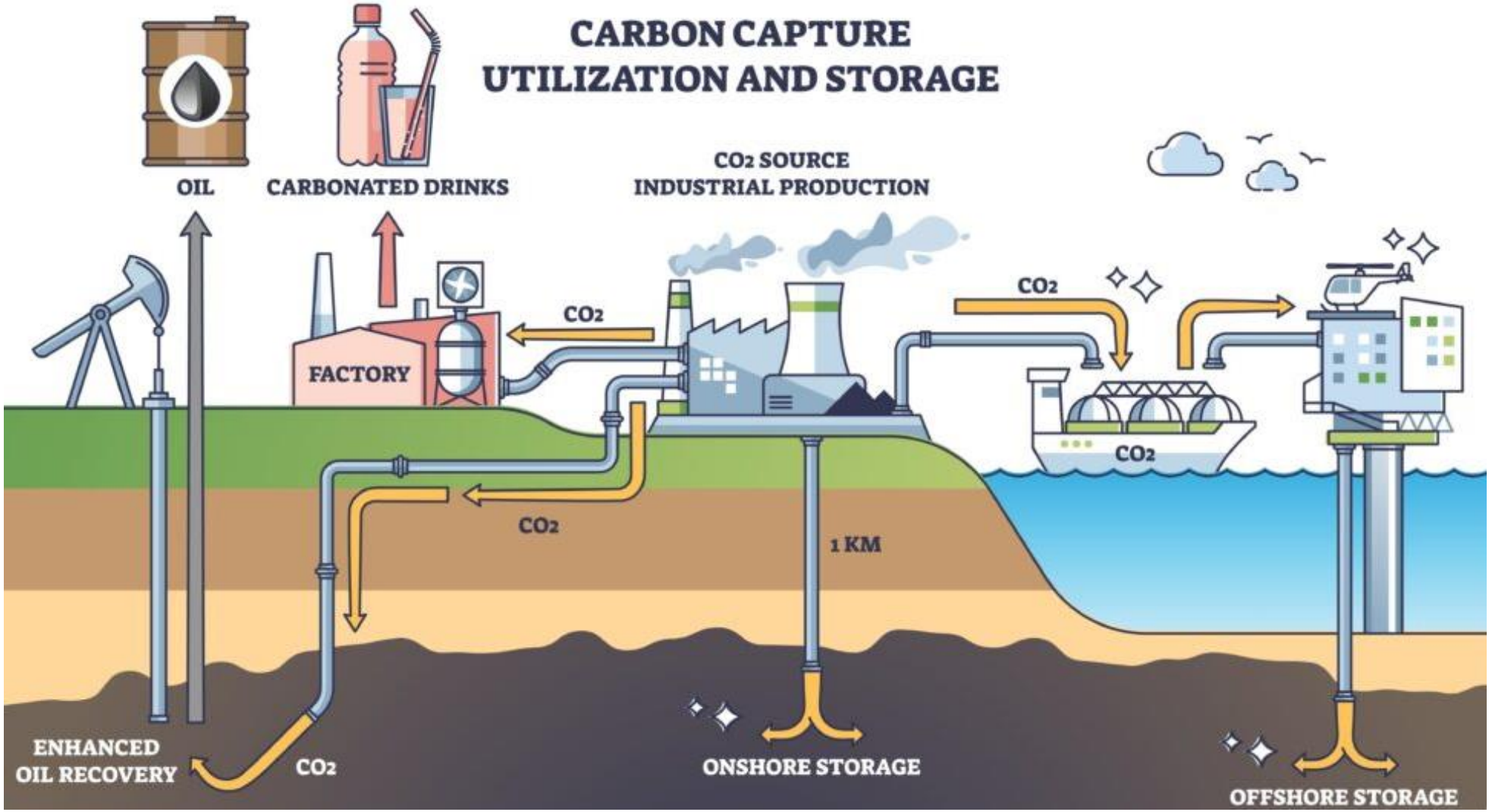
The climate crisis – why CCUS is essential to achieve Net Zero emissions



Professor Sophie Nixon
Department of Earth and Environmental Sciences
sophie.nixon@manchester.ac.uk; MIB

Focal areas of this lecture:

- Background context to the SDG ‘Climate Action’
- CCUS: What is it and why is it needed?
- CO_2 capture
- CO_2 utilisation
- CO_2 storage
- Paper of the Week

Intended Learning Outcomes:

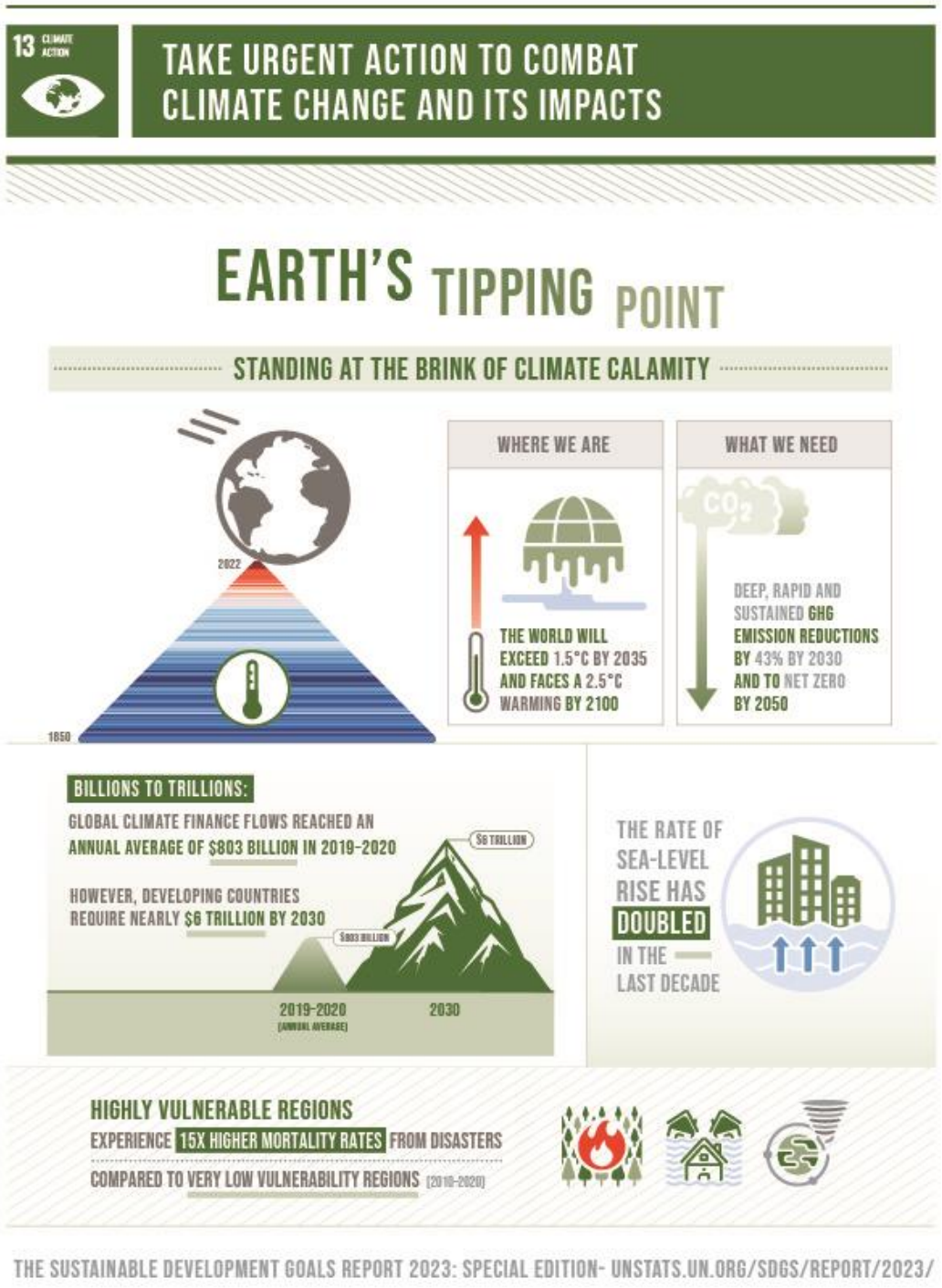
1. Know what CCUS is and what role it can play in achieving Net Zero emissions

2. Appreciate that CCUS is an interconnected toolbox

3. Recognise that many of these tools harness nature in some way (nature-based solutions)

4. Understand how the major capture / utilisation / storage approaches work

5. Appreciate the uncertainties of geological CO_2 storage

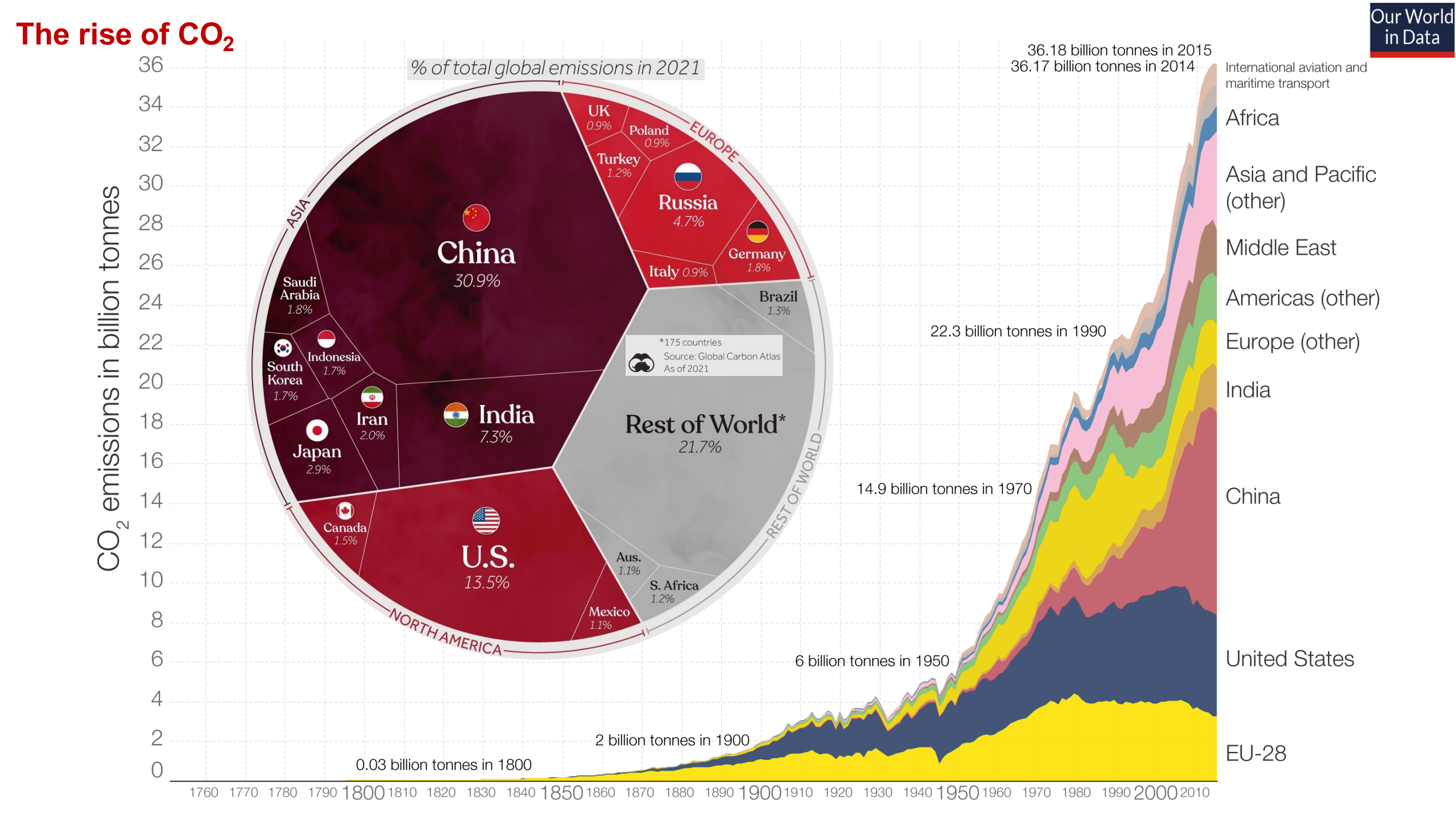


Key environmental science related challenges:

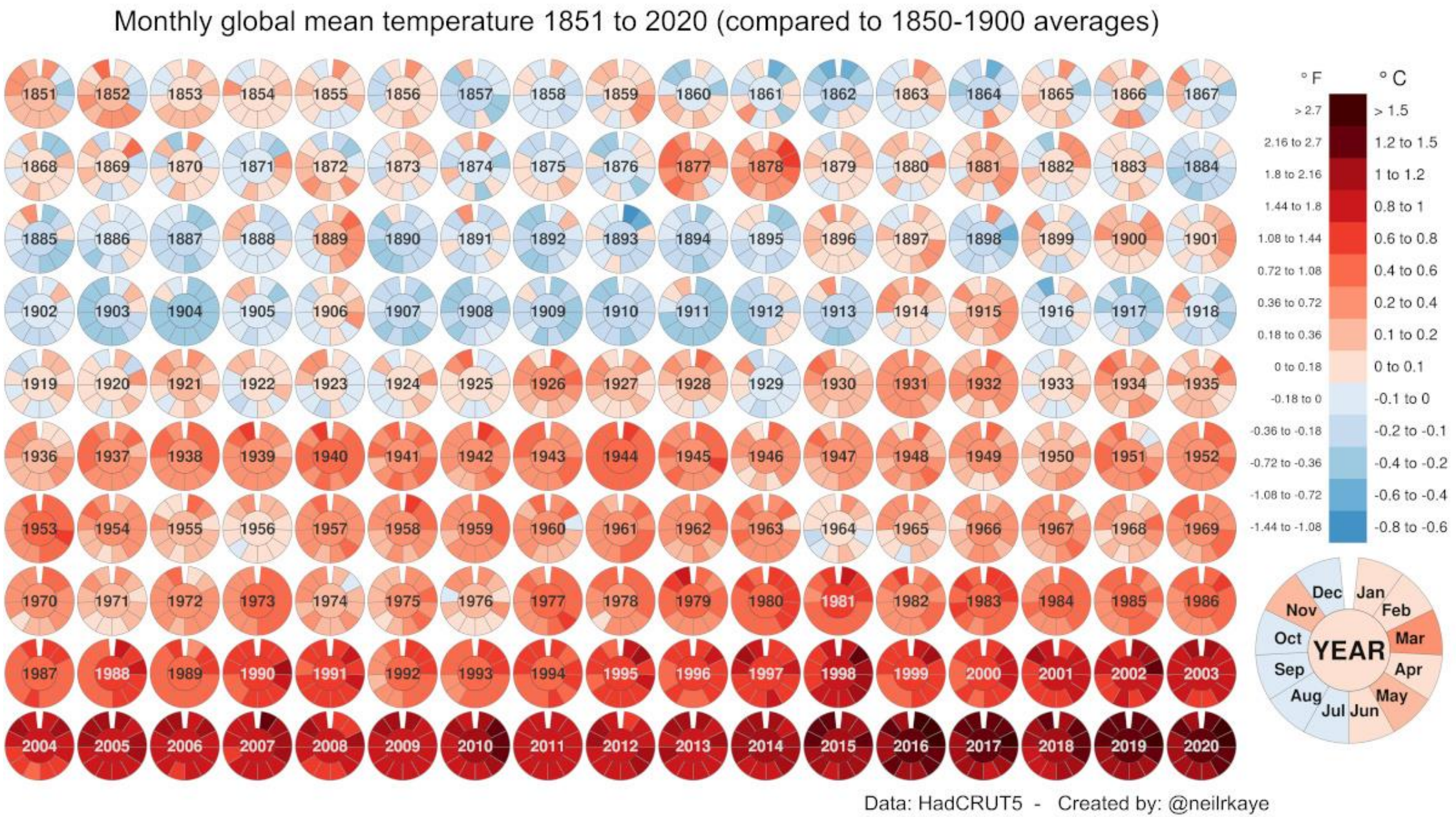
- Shift away from GHG-emitting processes, especially energy
- **Remember the soil!**
- **New technologies**: small changes in e.g. battery efficiency can make a big difference
- Multifactorial impacts still need to be quantified
- **Nature-based solutions**

2024 status: Marginal progress

The rise of CO_2



…and associated rise in global temperatures



GLOBAL Temperature Rise BY COUNTRY, 2022

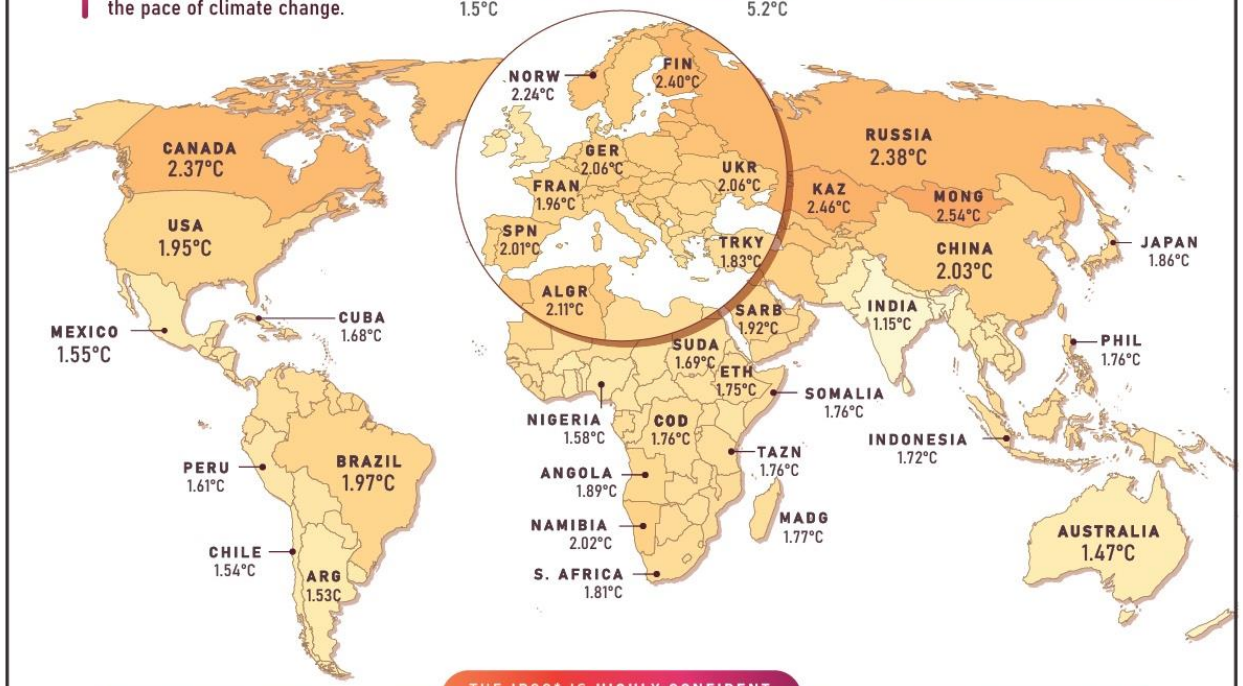
The Global North is generally warming the fastest due to Arctic amplification, where increased ice melt and feedback loops intensify the pace of climate change.

TEMPERATURE RISE
relative to average 1850-1900 temperatures

1.5°C — 5.2°C

NATIONAL AVERAGE, 2022
1.81°C

MAX	MIN
2.54°C	1.10°C



THE IPCC* IS HIGHLY CONFIDENT THAT THIS WARMING HAS:

- Led to more frequent and intense extreme weather events
- Caused irreversible ecosystem losses
- Globally reduced food and water security
- Adversely affected the physical and mental wellbeing of the global population

Source: Berkeley Earth, *Intergovernmental Panel on Climate Change

Projection data provided by BERKELEY EARTH.

GLOBAL Temperature Rise 2050 PROJECTIONS

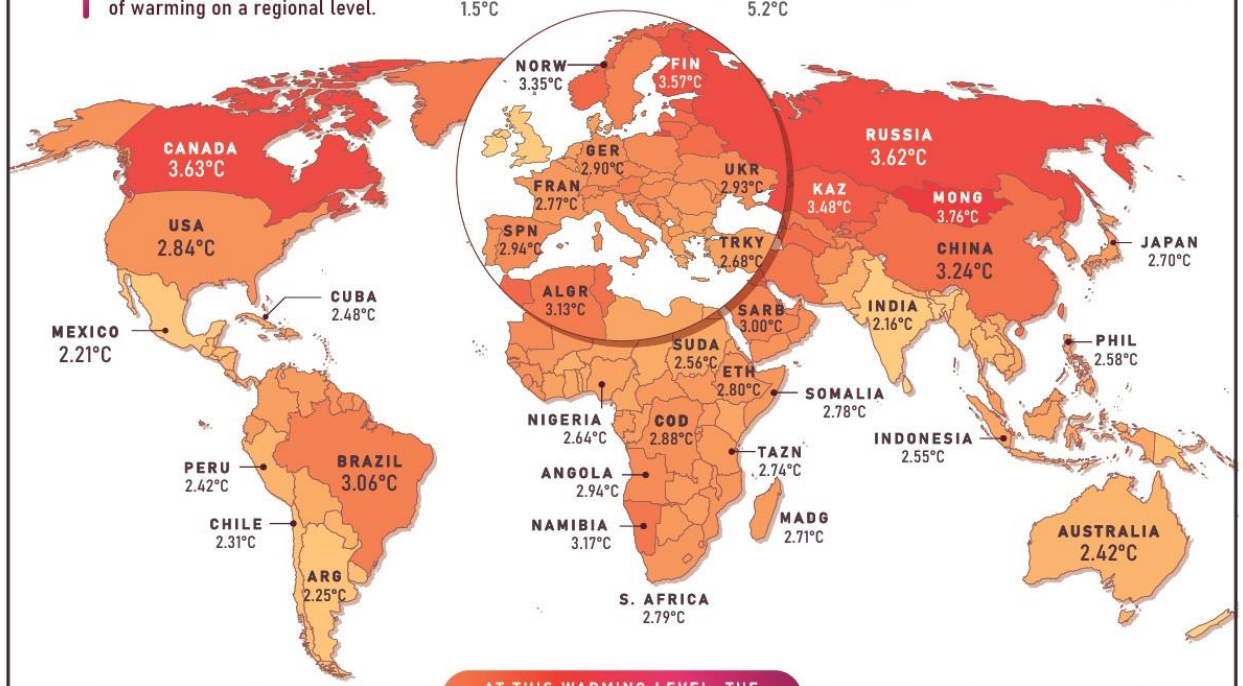
Arctic amplification, geography, weather patterns, ocean currents, and the influence of human activities all affect the level of warming on a regional level.

TEMPERATURE RISE
relative to average 1850-1900 temperatures

1.5°C — 5.2°C

NATIONAL AVERAGE, 2050
2.75°C

MAX	MIN
3.76°C	2.02°C



AT THIS WARMING LEVEL, THE IPCC IS HIGHLY CONFIDENT THAT:

- Extreme weather events will be more frequent and intense
- Nearly all ecosystems will face high risks of biodiversity loss
- Accelerated sea level rise will threaten coastal cities

Source: Berkeley Earth, Intergovernmental Panel on Climate Change

This map shows warming projections based on the IPCC's "middle-of-the-road" scenario (titled SSP2-4.5), which is the closest to expected emissions under current policies.

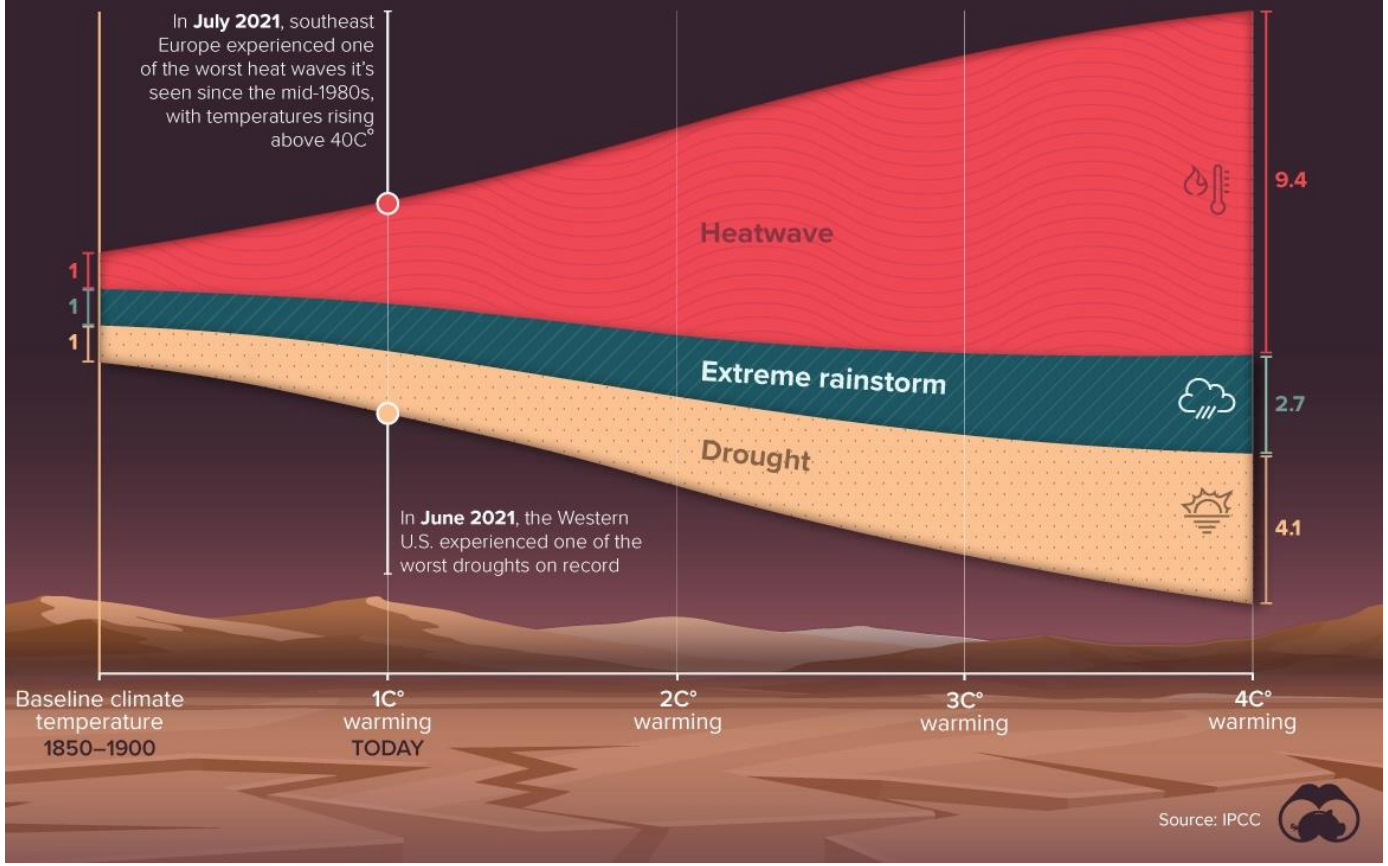
Projection data provided by BERKELEY EARTH.

The Accelerating Frequency of Extreme Climate Events

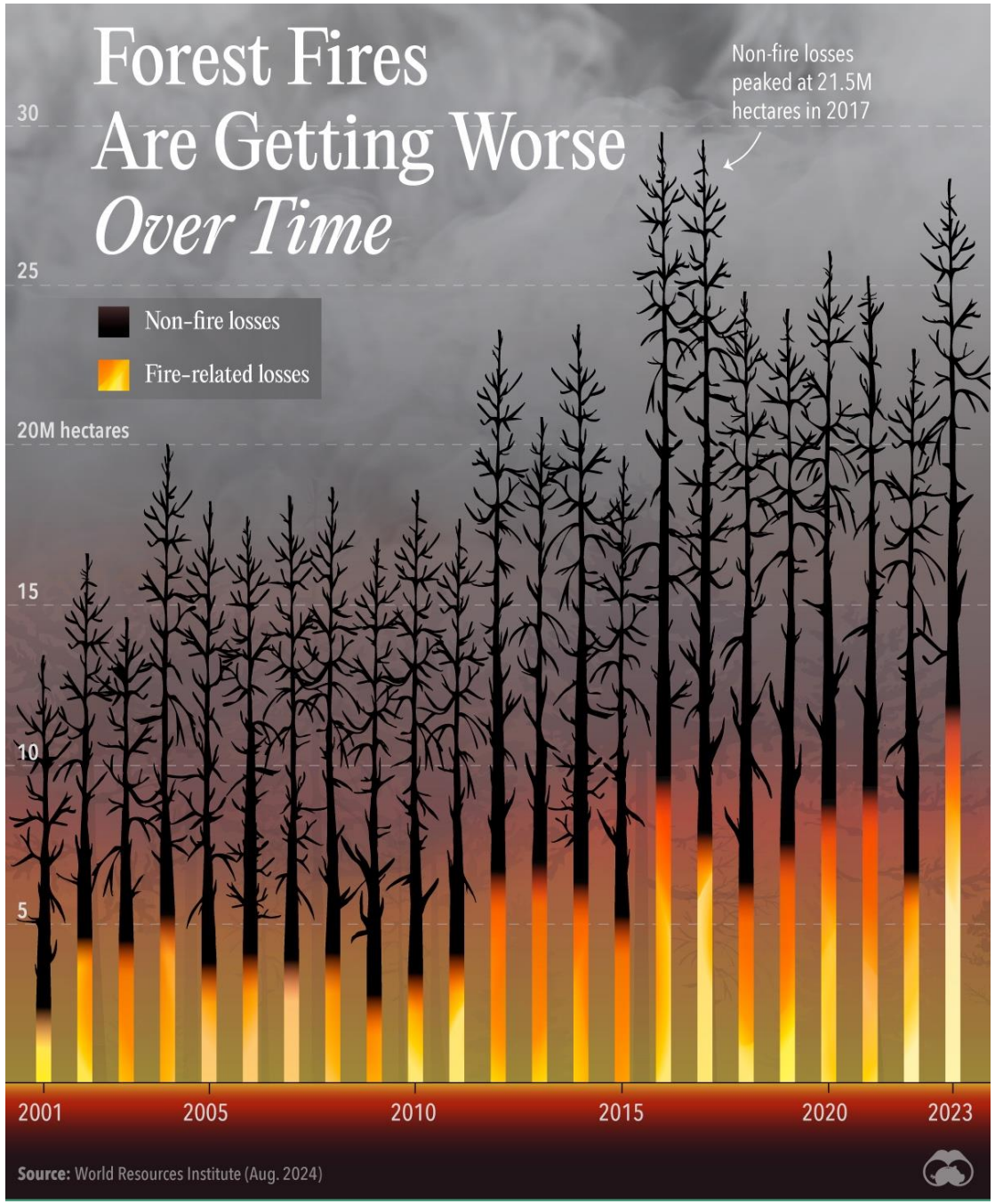
As the world heats up, extreme climate events like droughts and heatwaves are becoming increasingly more common. If temperatures continue to rise, things are expected to get even worse.

Here's how much the frequency of catastrophic climate events has increased in the past 200 years, and what could happen if temperatures keep rising.

Frequency of extreme events that used to be "once-in-a-decade"

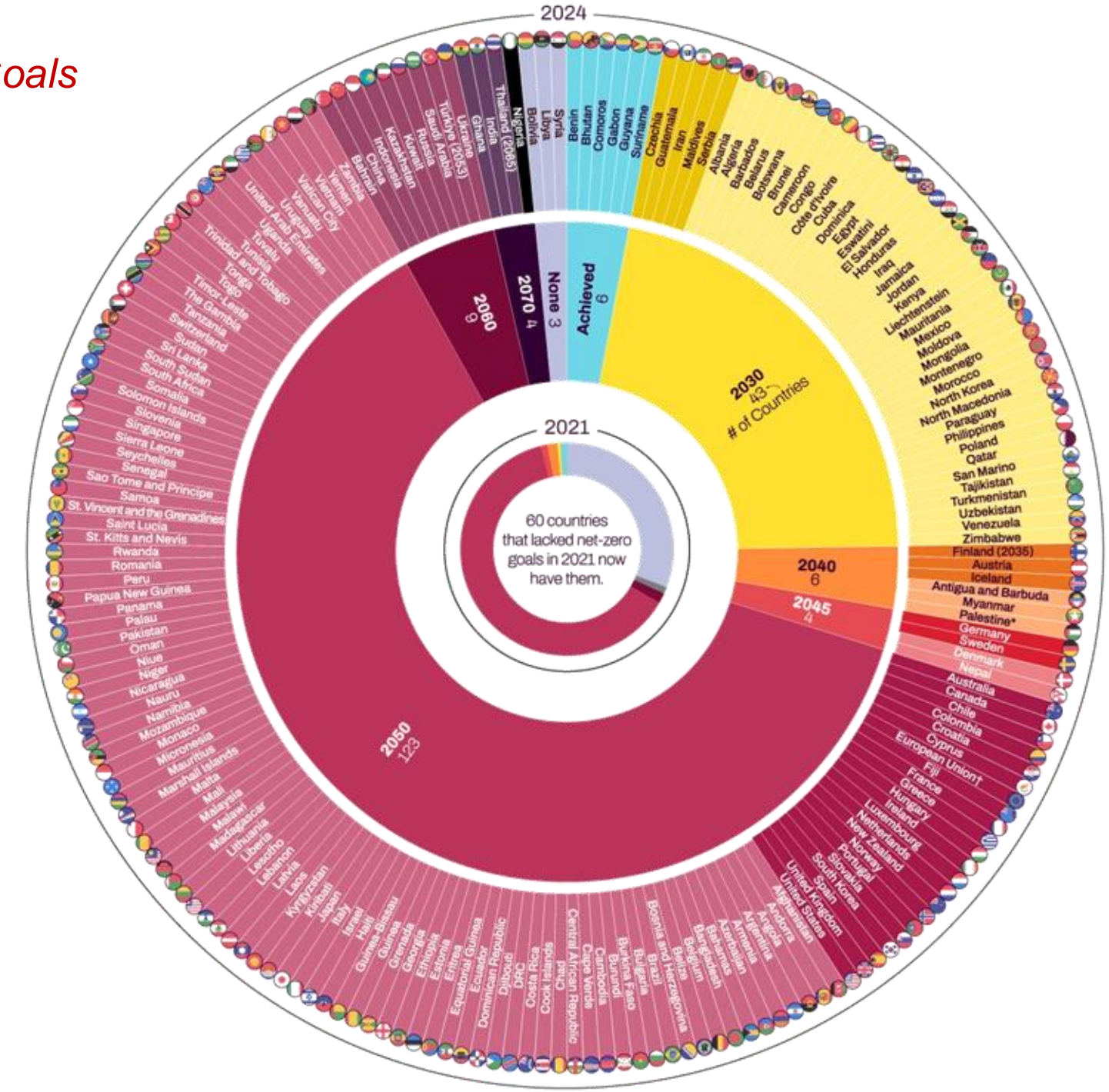


Forest Fires Are Getting Worse *Over Time*



The race to Net Zero

Emission Reduction Goals



Commitment		
Target year	In law	Not legally binding
2030		
2040		
2045		
2050		
2060		
2070		
2050 -2100		
	Achieved	
	None	

2045
Germany
Sweden
Denmark

2050
Australia
Canada
France
Netherlands
New Zealand
Norway
UK
USA???????

2060
China
Russia

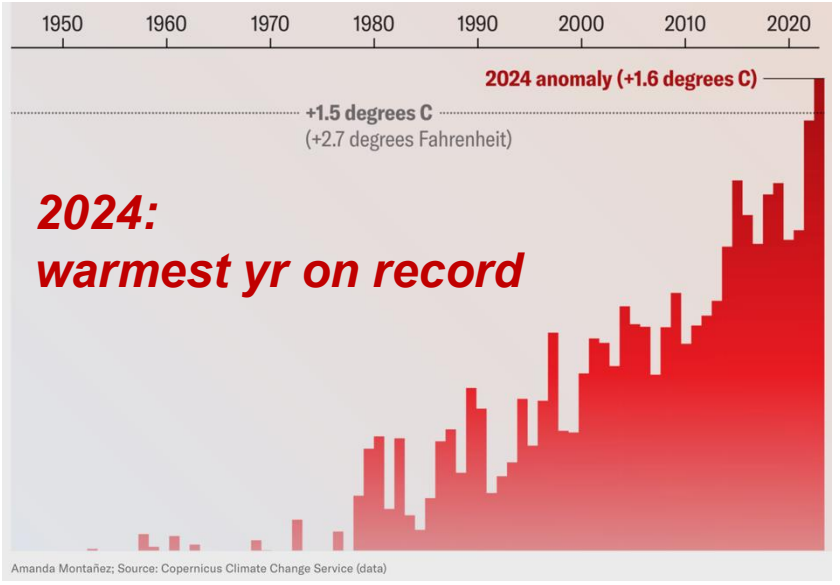
2070
India

VISUAL CAPITALIST

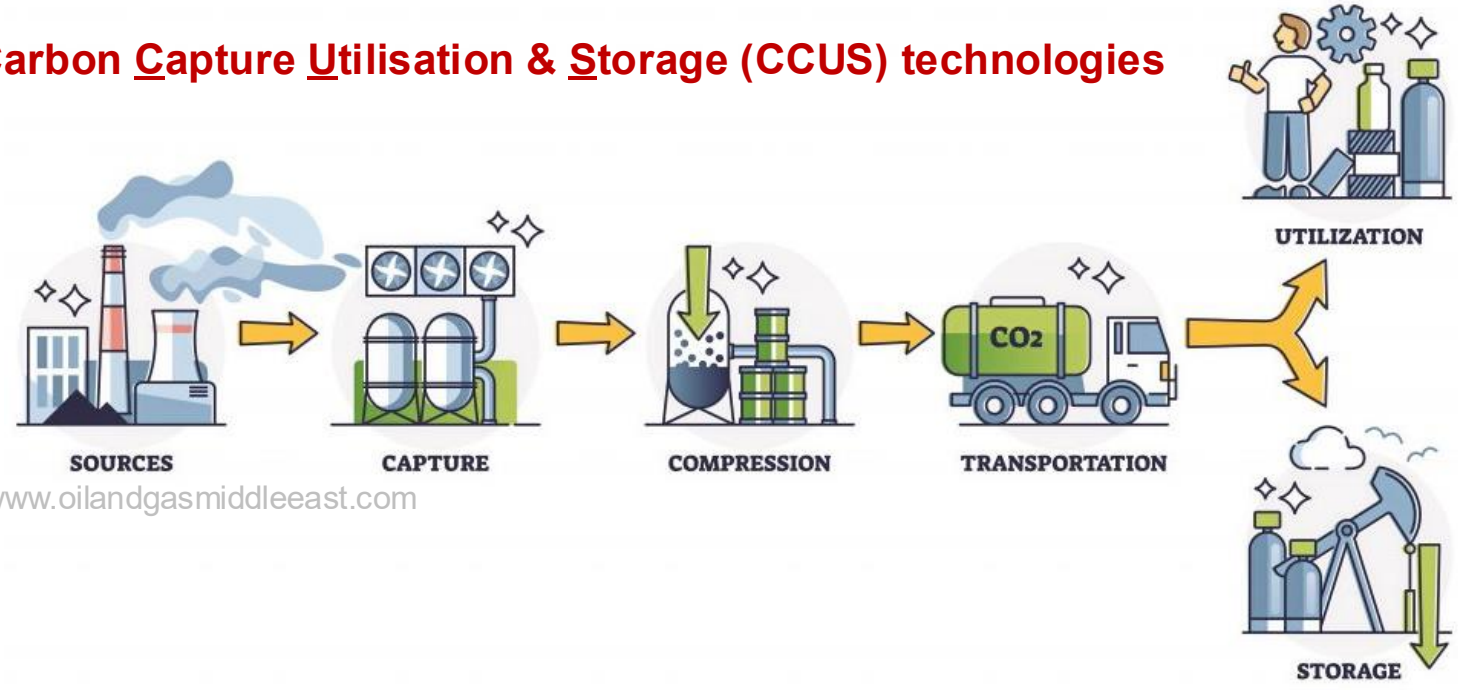
Net Zero CO_2 emissions is an urgent yet distant goal

"Our world needs climate action on all fronts: ***everything, everywhere, all at once****."*

UN secretary general, António Guterres



Carbon Capture Utilisation & Storage (CCUS) technologies



www.oilandgasmiddleeast.com

Trump victory is a major setback for climate action, experts say



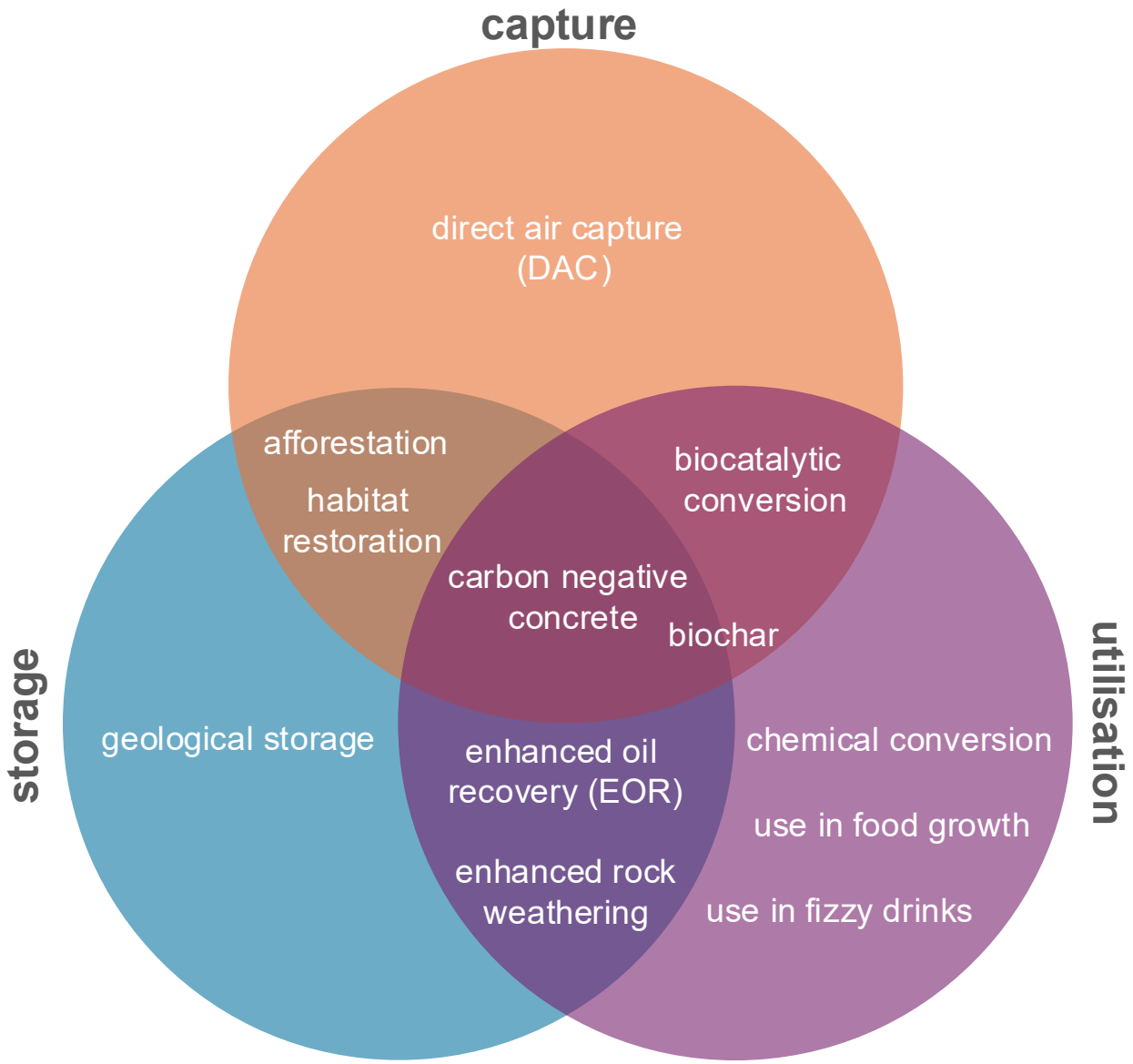
Matt McGrath ›
Environment correspondent
@mattmcgrathbbc ›

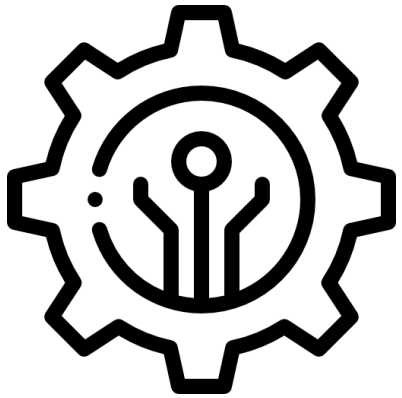
"drill, baby, drill"

CCUS is an interconnected toolbox



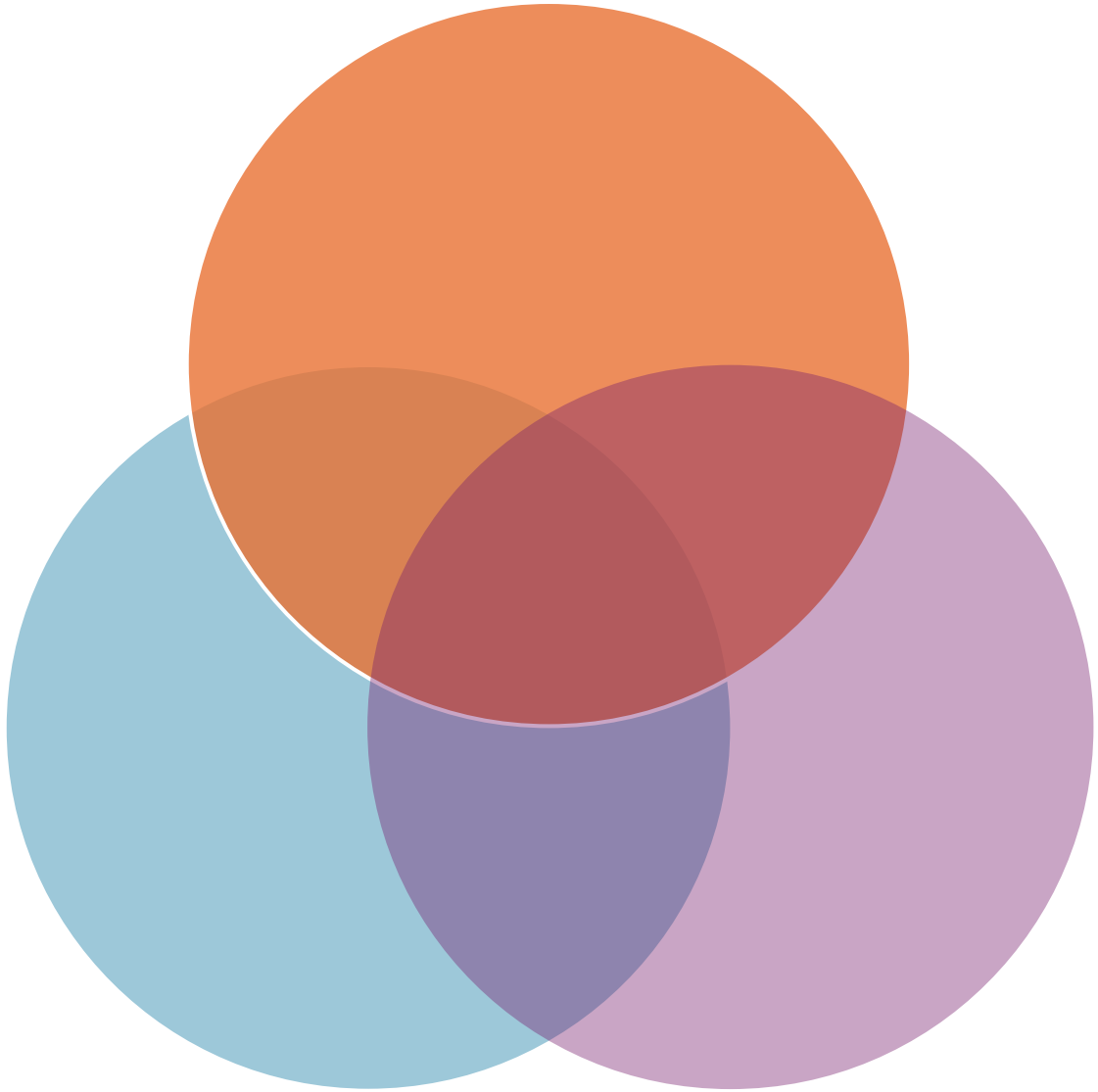
nature-based solutions



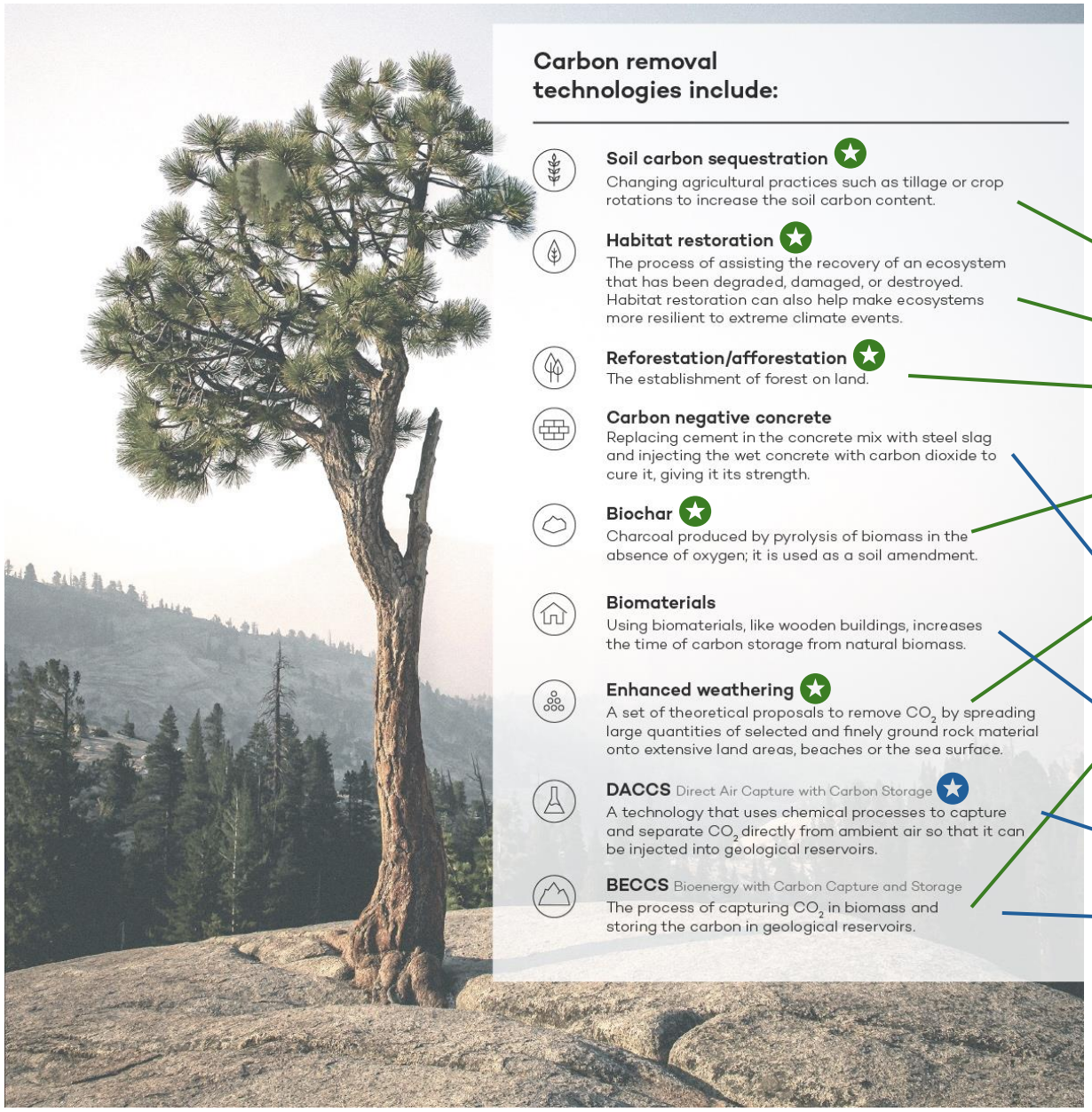


disruptive technologies

CO_2 Capture

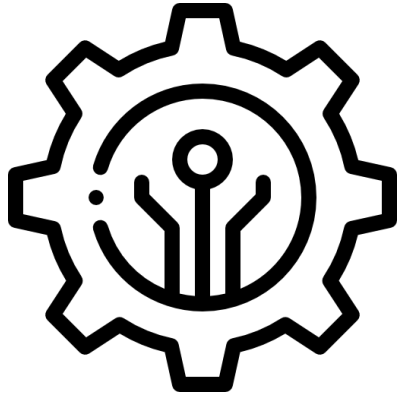


Carbon Capture (CCUS)



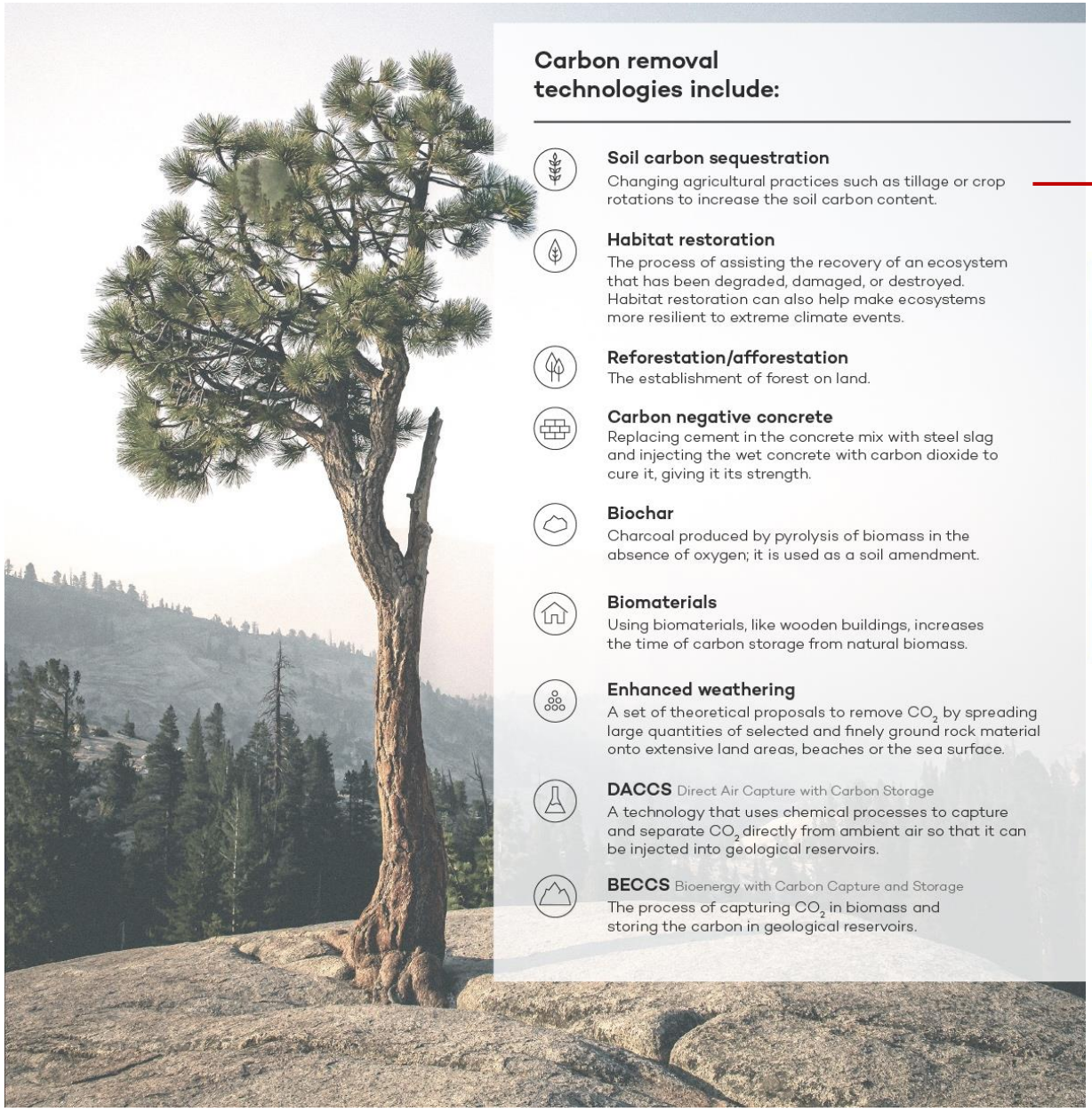


nature-based solutions

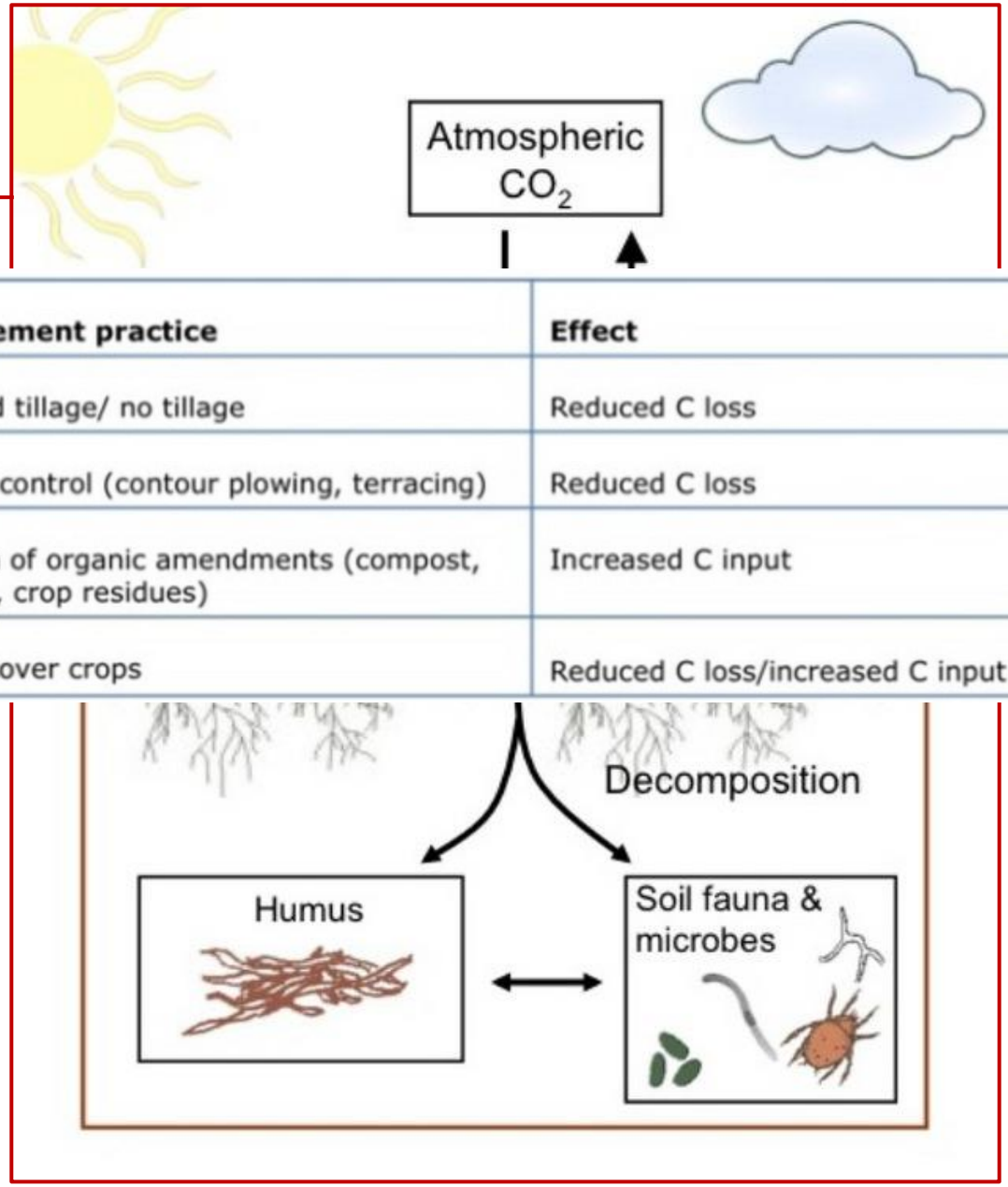


disruptive technologies

Carbon Capture (CCUS)



soil carbon sequestration

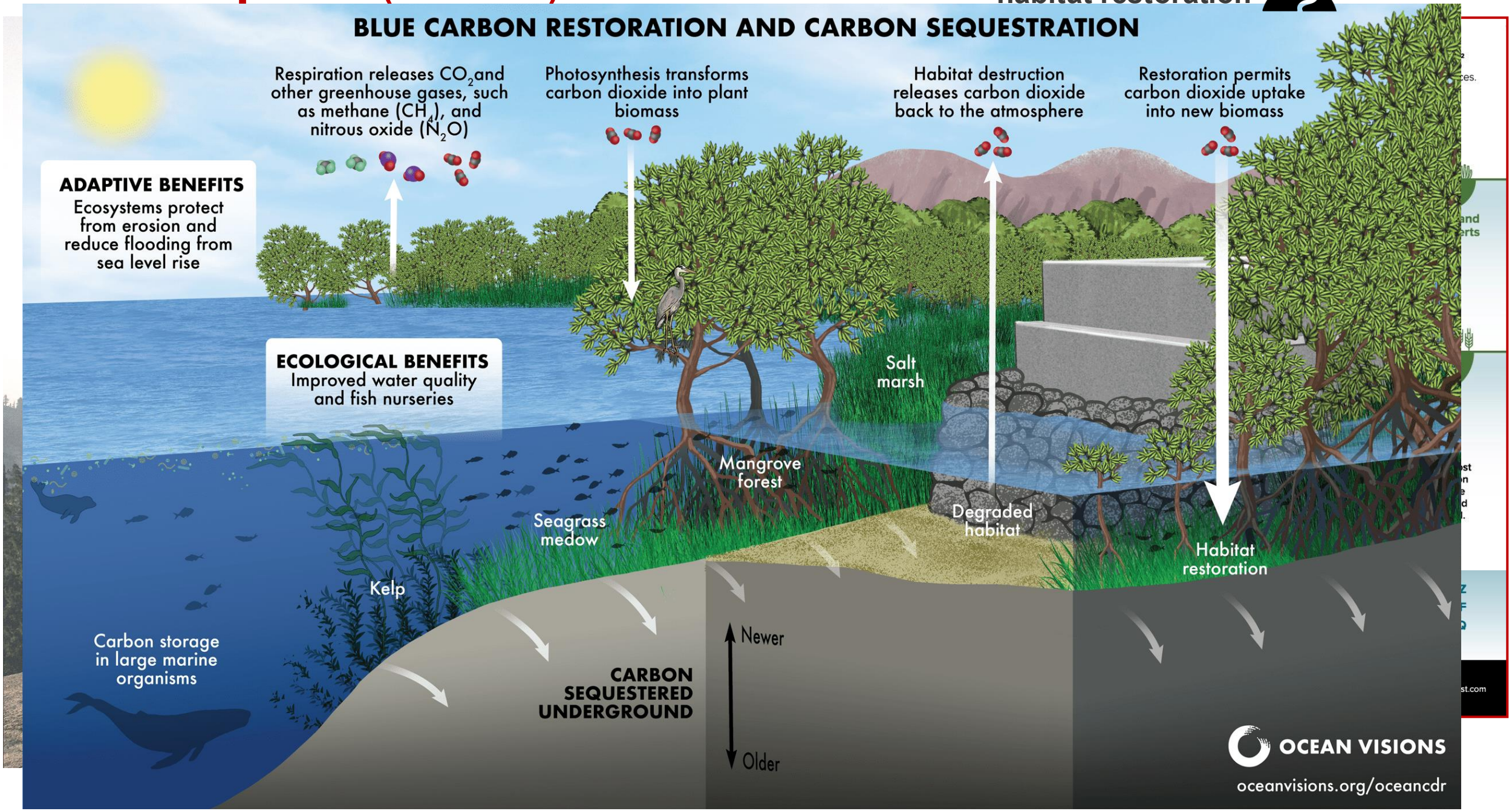


Management practice	Effect
Reduced tillage/ no tillage	Reduced C loss
Erosion control (contour plowing, terracing)	Reduced C loss
Addition of organic amendments (compost, manure, crop residues)	Increased C input
Use of cover crops	Reduced C loss/increased C input

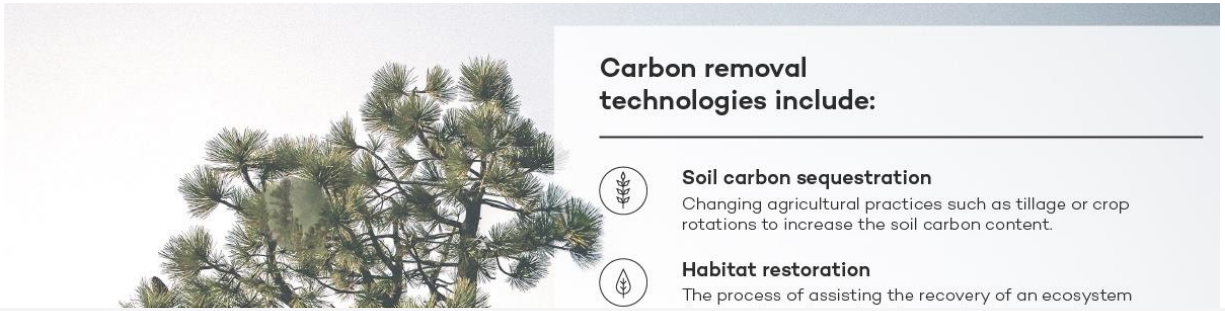
Ontl & Schulte (2012) Soil Carbon Storage. *Nature Education Knowledge* 3(10):35

Carbon Capture (CCUS)

habitat restoration



Carbon Capture (CCUS)



The below table ranks the rest of the houseplants in the top ten, compared to the prayer plant:

Type of Plant	Reduction of CO_2 (Percentage)	CO_2 Absorbed (in kg)
Prayer Plant	14.40%	0.0002677 kg of CO_2 absorbed
Rubber Plant	12.84%	0.0002387 kg of CO_2 absorbed
Bird's Nest Fern	12.48%	0.0002320 kg of CO_2 absorbed
Dieffenbachia	11.10%	0.0002063 kg of CO_2 absorbed
Anthurium	10.80%	0.0002008 kg of CO_2 absorbed
Arrowhead Plant	10.08%	0.0001874 kg of CO_2 absorbed
Peace Lily	10.05%	0.0001868 kg of CO_2 absorbed
Golden Pothos	10.03%	0.0001865 kg of CO_2 absorbed
Yucca	3.60%	0.0000669 kg of CO_2 absorbed
Spider Plant	0.10%	0.0000019 kg of CO_2 absorbed[14]



reforestation/afforestation



8billiontrees.com





Return Distance Miles	Carbon Emissions (in tons)
12,318	2,96 tons of CO_2 emissions
12,308	2.96 tons of CO_2 emissions
10,458	2.52 tons of CO_2 emissions
107,36	2.59 tons of CO_2 emissions

Carbon Capture (CCUS)

biochar



Carbon removal technologies include:

Soil carbon sequestration
Changing agricultural practices such as tillage or crop rotations to increase the soil carbon content.

Habitat restoration
The process of assisting the recovery of an ecosystem that has been degraded, damaged, or destroyed. Habitat restoration can also help make ecosystems more resilient to extreme climate events.

Reforestation/afforestation
The establishment of forest on land.

Carbon negative concrete
Replacing cement in the concrete mix with steel slag and injecting the wet concrete with carbon dioxide to cure it, giving it its strength.

Biochar
Charcoal produced by pyrolysis of biomass in the absence of oxygen; it is used as a soil amendment.

Biomaterials
Using biomaterials, like wooden buildings, increases the time of carbon storage from natural biomass.

Enhanced weathering
A set of theoretical proposals to remove CO_2 by spreading large quantities of selected and finely ground rock material onto extensive land areas, beaches or the sea surface.

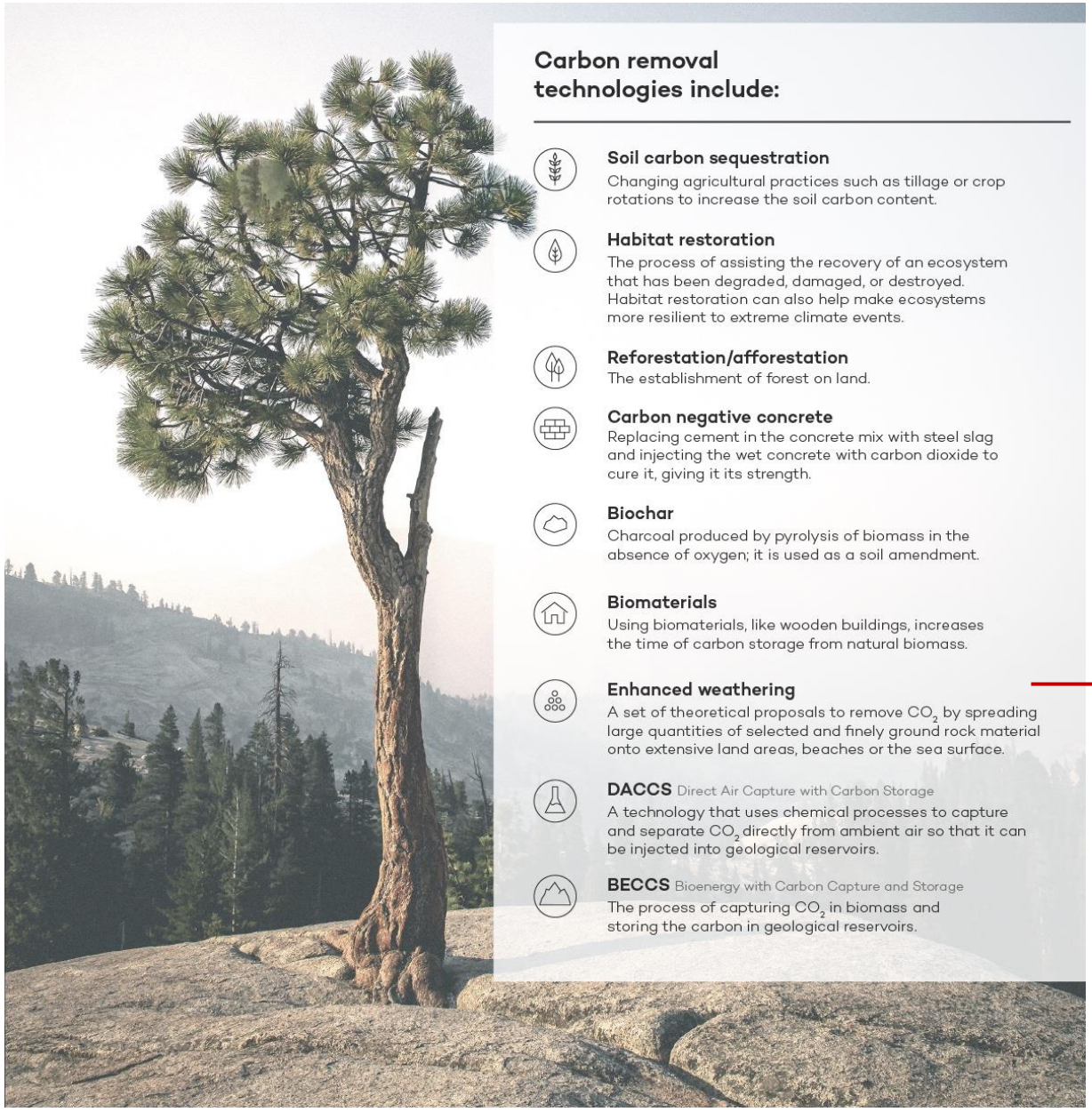
DACCS Direct Air Capture with Carbon Storage
A technology that uses chemical processes to capture and separate CO_2 directly from ambient air so that it can be injected into geological reservoirs.

BECCS Bioenergy with Carbon Capture and Storage
The process of capturing CO_2 in biomass and storing the carbon in geological reservoirs.



Al Masud *et al* (2023) *Science of The Total Environment* **904**:166813

Carbon Capture (CCUS)



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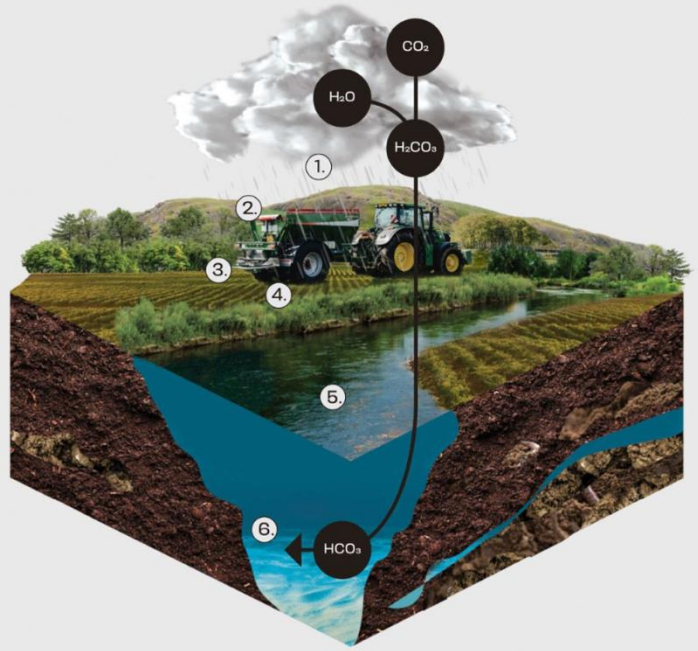
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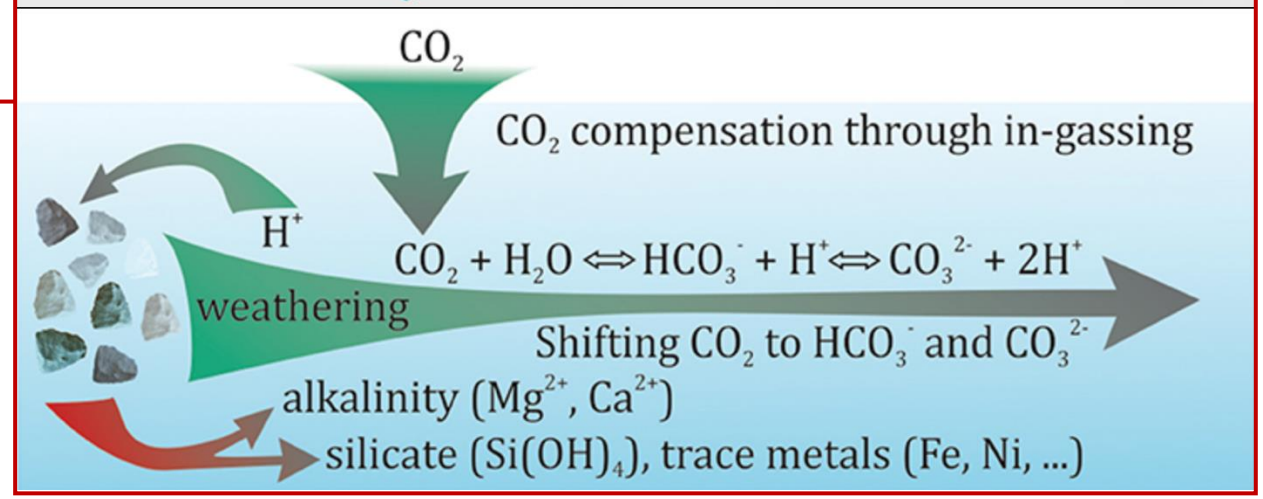
BECCS Bioenergy with Carbon Capture and Storage
The process of capturing CO_2 in biomass and storing the carbon in geological reservoirs.

enhanced rock weathering



ENHANCED ROCK WEATHERING

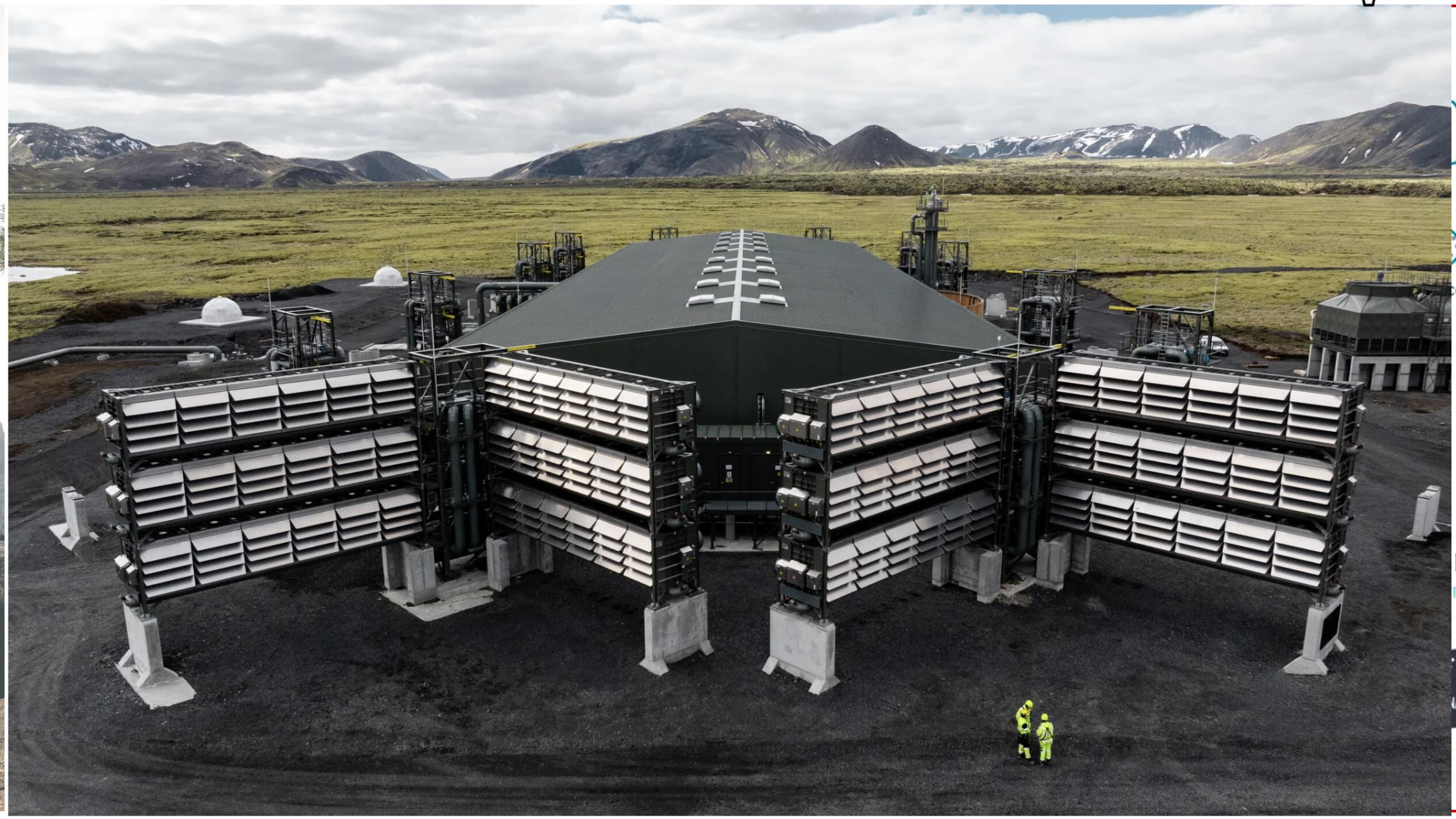
1. Rain combines with atmospheric CO_2 to form a dilute carbonic acid (H_2CO_3)
2. UNDO spreads crushed silicate rock on agricultural land
3. Rainfall reacts with the rock, capturing CO_2 in bicarbonate form (HCO_3)
4. Nutrients from the rock improve soil health and crop yield
5. Bicarbonates wash into rivers and fall to the ocean floor, permanently locking away CO_2
6. Alkaline bicarbonates combat ocean acidification and support shell formation



Bach et al (2019) *Frontiers in Climate* 1:DOI=10.3389/fclim.2019.00007

Carbon Capture (CCUS)

direct air capture (DAC)



Mammoth, the world's largest direct air capture and storage plant, is designed for a nameplate capture capacity of up to 36,000 tons of CO_2 per year.

Carbon Capture (CCUS)

Smithsonian MAGAZINE

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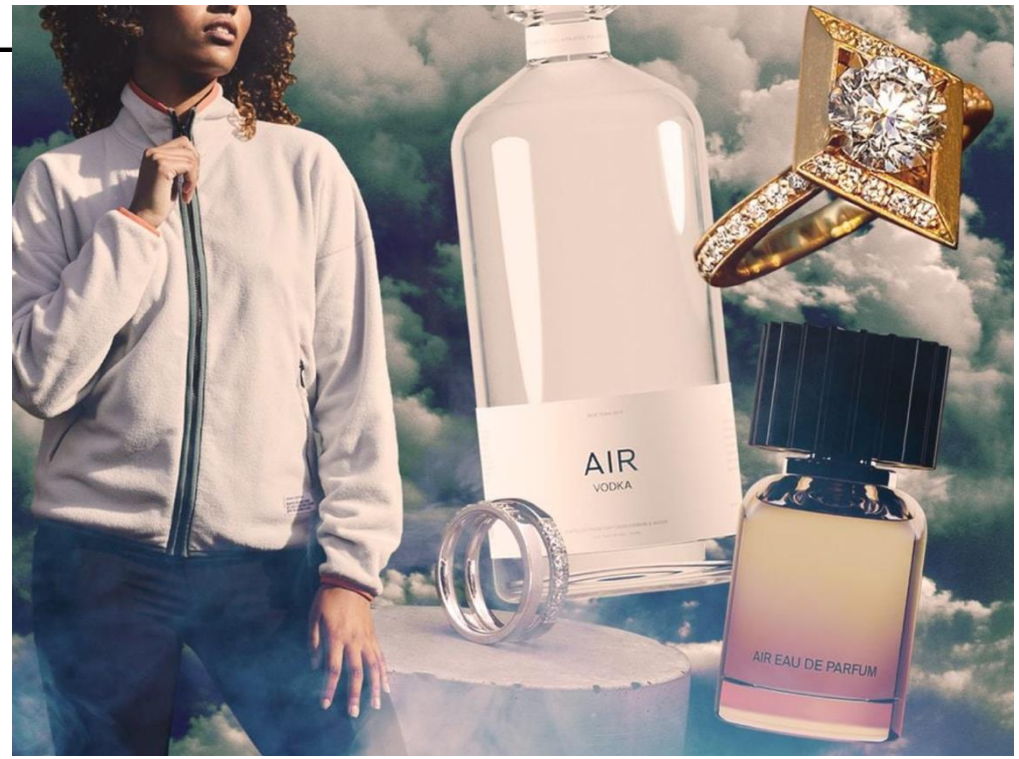
INNOVATION

Little Luxuries Made With Captured Pollution Hint at Big Frontiers in Climate Science

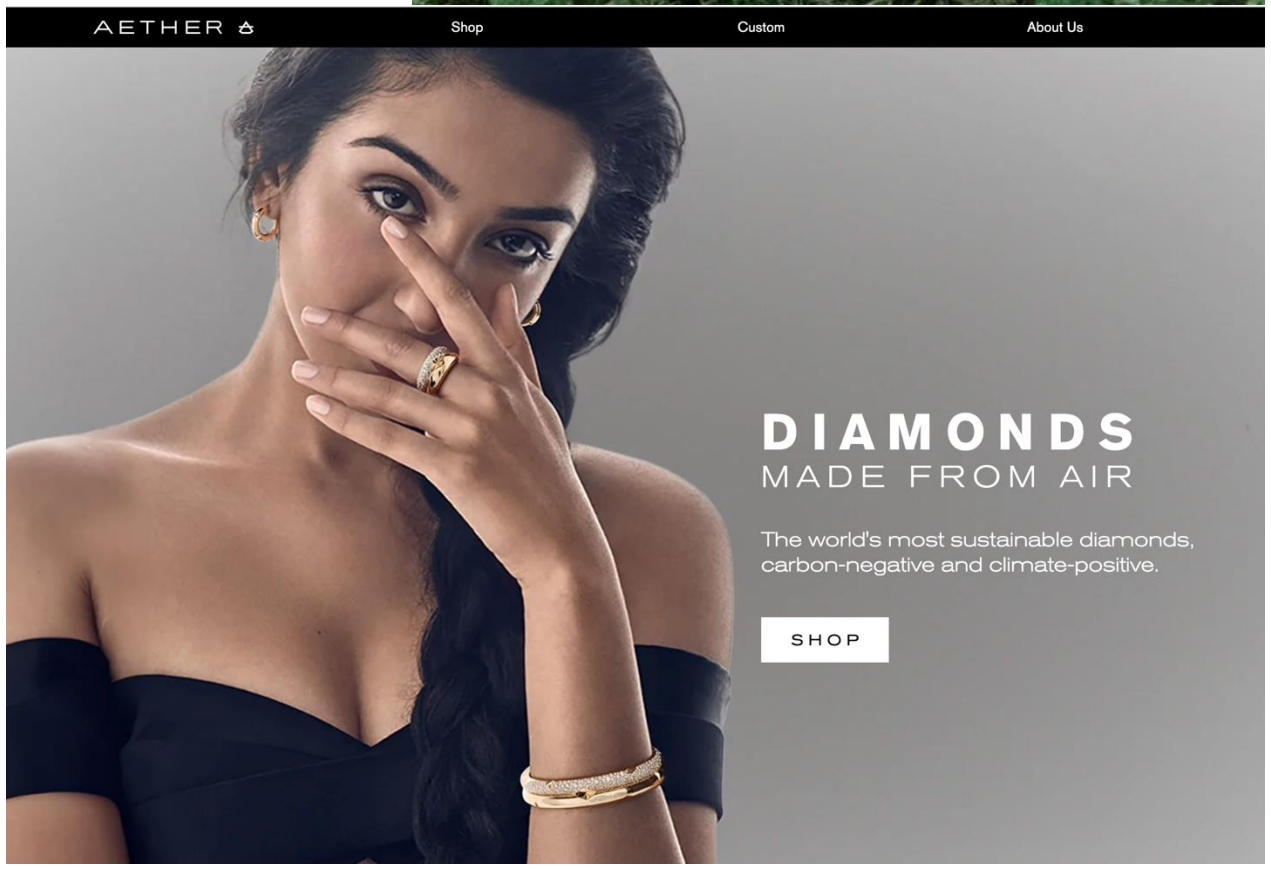
Entrepreneurs are using jewelry, fragrances and clothing to demonstrate what's possible with repurposed carbon—and environmentalists have questions

Lauren Oster

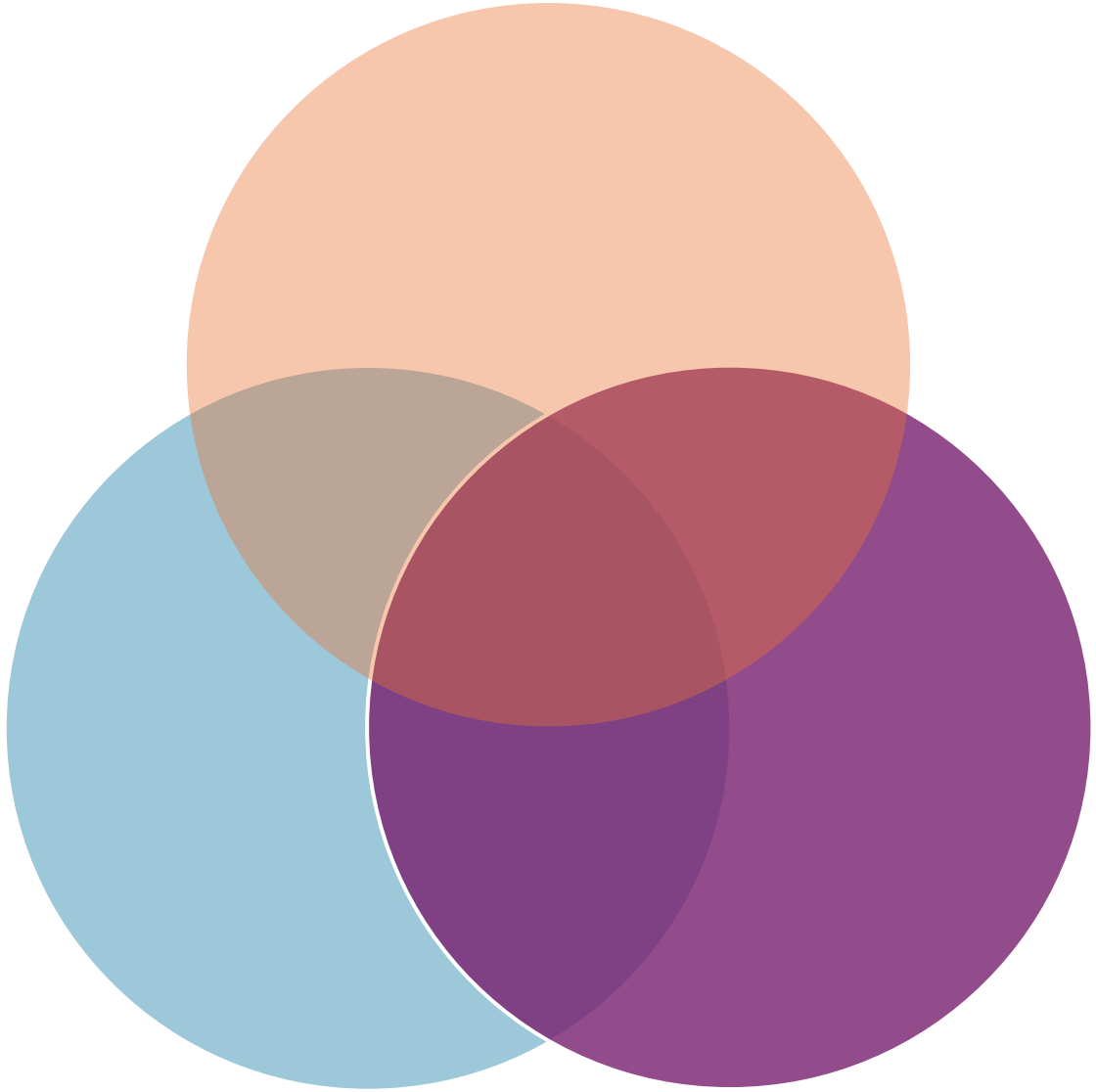
September 21, 2023







CO_2 Utilisation



CO_2 Utilisation (CCUS)

Perspective | Published: 06 November 2019

The technological and economic prospects for CO_2 utilization and removal

Cameron Hepburn, Ella Adlen ✉, John Beddington, Emily A. Carter, Sabine Fuss, Niall Mac Dowell, Jan C. Minx, Pete Smith & Charlotte K. Williams

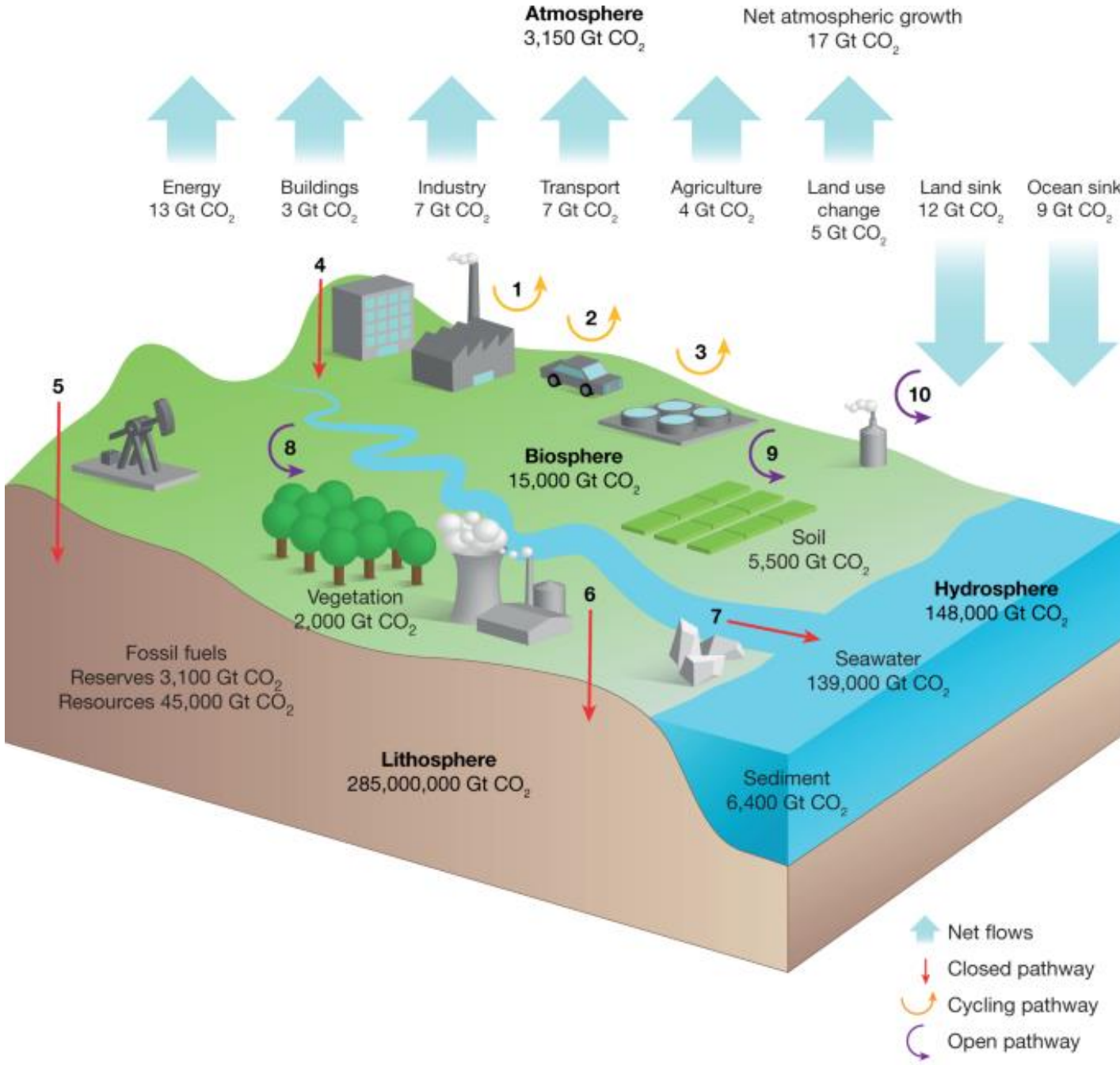
Nature **575**, 87–97 (2019) | Cite this article

169k Accesses | **401** Altmetric | Metrics

"CO_2 utilization does not necessarily reduce emissions and does not necessarily deliver a net climate benefit, once indirect and other effects have been accounted for."

Pathway

(1) Chemicals from CO_2
(2) Fuels from CO_2
(3) Products from microalgae
(4) Concrete building materials
(5) CO_2-EOR
(6) Bioenergy with carbon capture and storage
(7) Enhanced weathering
(8) Forestry techniques
(9) Soil carbon sequestration techniques
(10) Biochar



CO_2 Utilisation (CCUS)

VISOLIS

Advanced Data Modeling

AI POWERED ENGINE

Precision Biology

ENGINEER EFFICIENT PRODUCTION MICROBE

Process Development & Scale Up

SCALE FROM LAB TO COMMERCIAL



High performance materials

Using synthetic biology, we program our microorganisms to use a wide range of renewable feedstocks instead of petroleum.

The outcome is an identical molecule that can be used today with net-zero or negative emissions.

Synthetic Biology

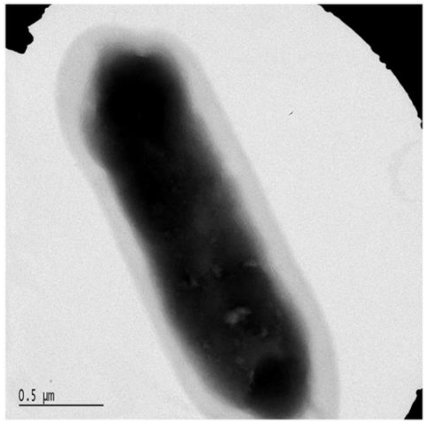
Harnessing the
dioxide and hy

We're starting with protein, but have no plans on stopping there.

BIOMASON

CO_2 Utilisation (CCUS)

LanzaTech



BRIEF RESEARCH REPORT article

Front. Bioeng. Biotechnol. , 27 March 2020

Sec. Bioprocess Engineering

Volume 8 - 2020 | https://doi.org/10.3389/fbioe.2020.00204

This article is part of the Research Topic
Recent Advances in Continuous Cultivation

View all 10 articles >

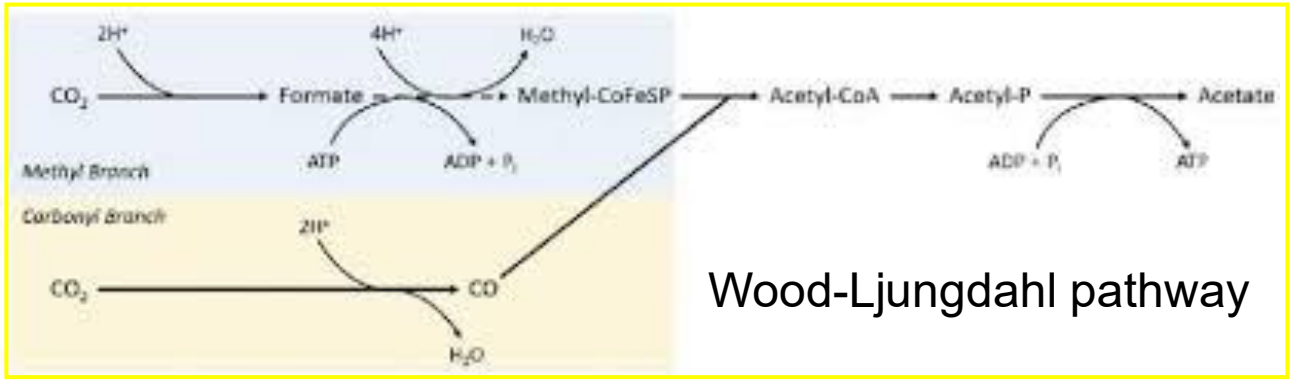
Enhancing CO_2-Valorization Using *Clostridium autoethanogenum* for Sustainable Fuel and Chemicals Production

James K. Heffernan[1†] Kaspar Valgepea[1,2†] Renato de Souza Pinto Lemgruber[1]

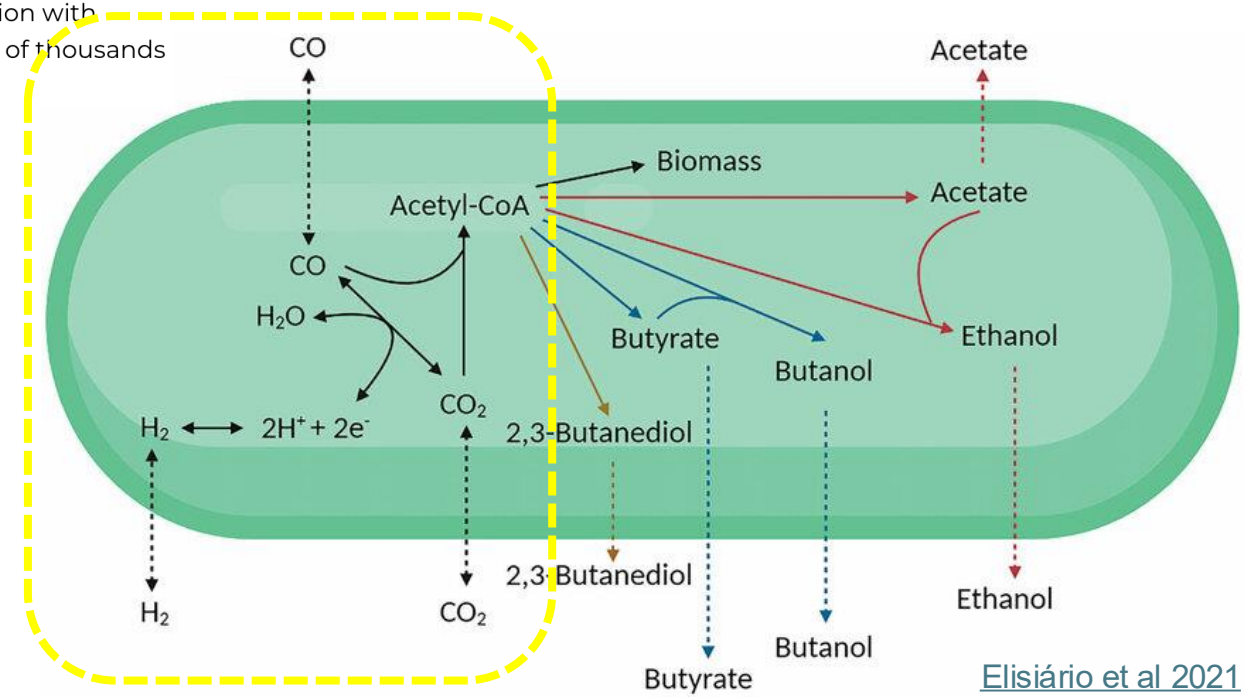
Isabella Casini[3] Manuel Plan[4] Ryan Tappel[5] Sean D. Simpson[5]

Michael Köpke[5] Lars K. Nielsen[1,4] Esteban Marcellin[1,4*]

At LanzaTech, we've partnered with some of the smallest living beings on Earth to build a climate solution with gigaton-scale potential. Those living things are called microbes, and they're helping us abate hundreds of thousands of tonnes of planet-warming carbon emissions each year.



Wood-Ljungdahl pathway



Elisiário et al 2021

CO_2 Utilisation (CCUS)

C3 BIOTECH

Advanced Synthetic Fuels

KEROSENE

C3 BIOTECH has partnered with a US Centre to develop bio-based production routes for aviation fuel precursors at scale.

Biotechnol Biofuels Bioprod. 2023 Oct 11;16(1):152. doi: 10.1186/s13068-023-02404-1.

Chemoautotrophic production of gaseous hydrocarbons, bioplastics and osmolytes by a novel Halomonas species

Matthew Faulkner [1], Robin Hoeven [2 3], Paul P Kelly [1], Yaqi Sun [4], Helen Park [1], Lu-Ning Liu [4], Helen S Toogood [5 6], Nigel S Scrutton [7 8]

Affiliations + expand

PMID: 37821908 PMCID: PMC10568851 DOI: 10.1186/s13068-023-02404-1

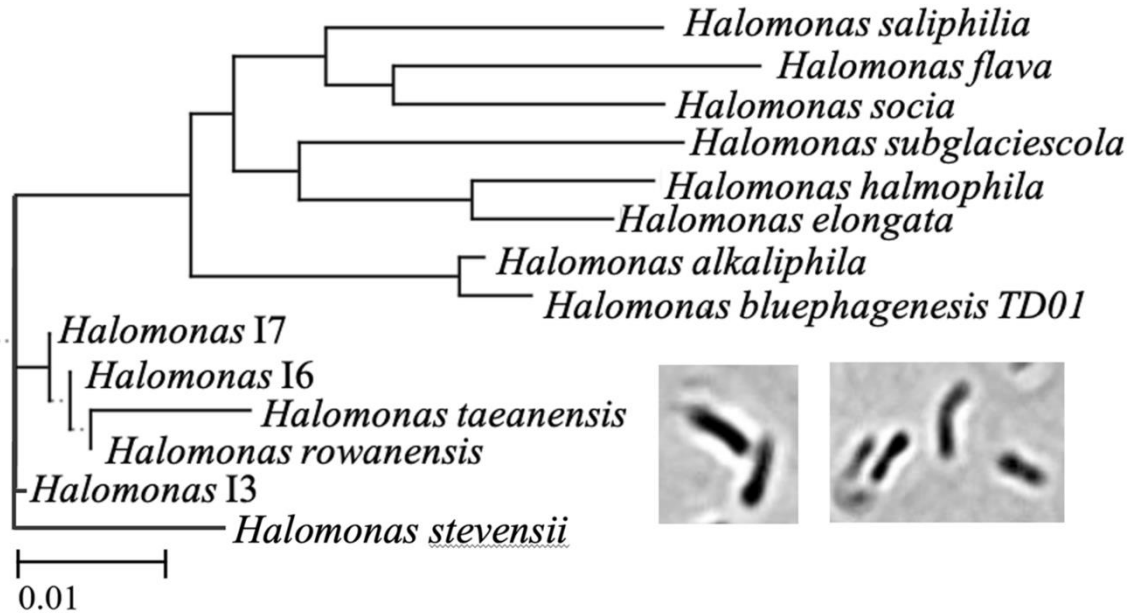
ELSEVIER

Current Opinion in Biotechnology

Volume 85, February 2024, 103064

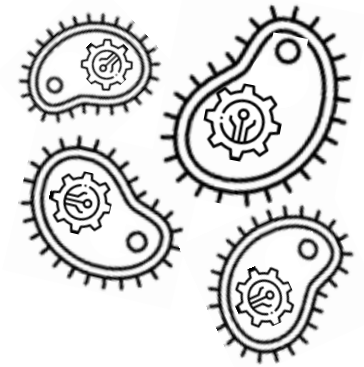
Nonsterile microbial production of chemicals based on *Halomonas* spp

Jing Zhang [1 2 #], Xu Yan [3 #], Helen Park [3 4], Nigel S Scrutton [4], Tao Chen [1 2 ✉], Guo-Qiang Chen [3 5 6 ✉]



CO_2 Utilisation (CCUS)

nizon lab



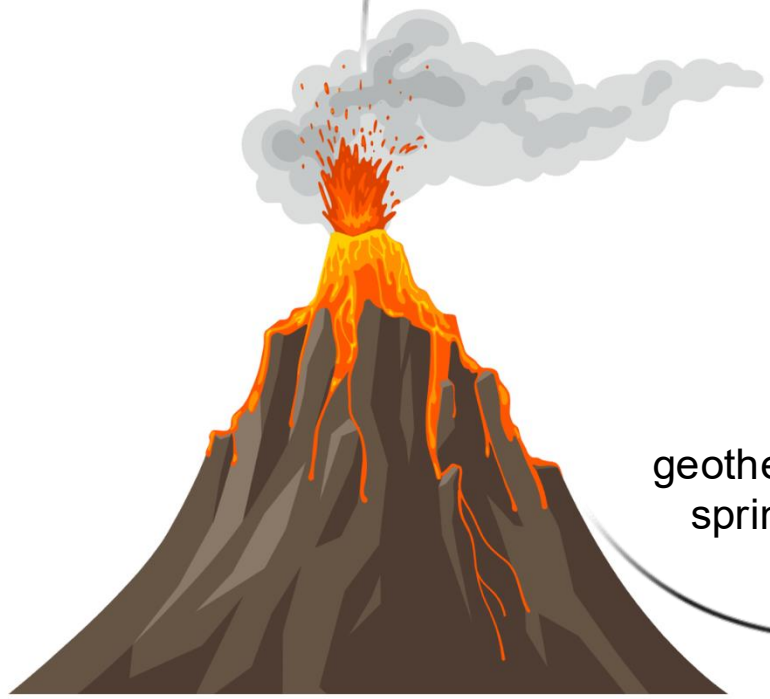
***CO_2 bioconversion independent of light by microbial communities** (microbiomes)*

volcanic outgassing

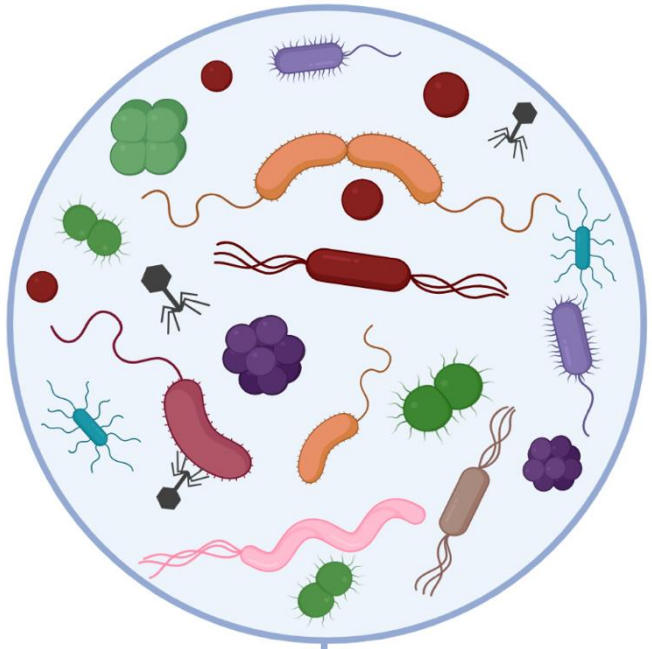
CO_2

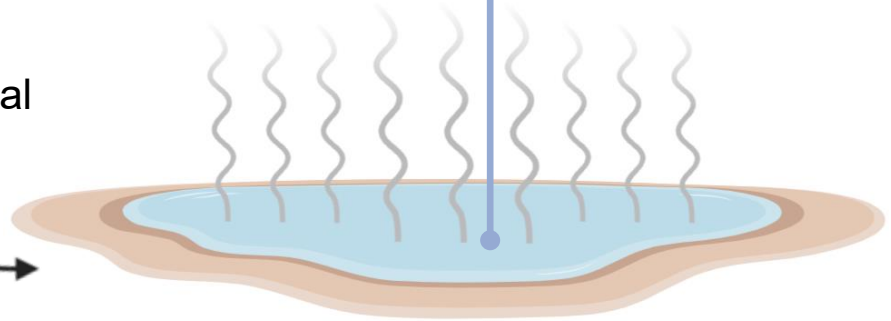
H_2

nutrients

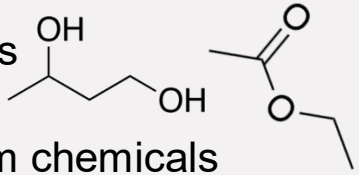


geothermal springs





biofuels



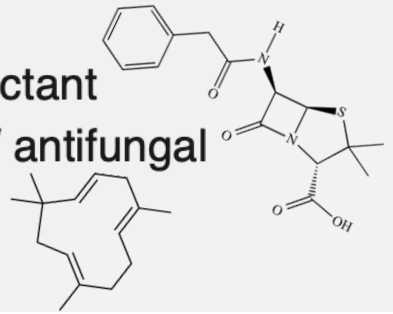
platform chemicals

pigment

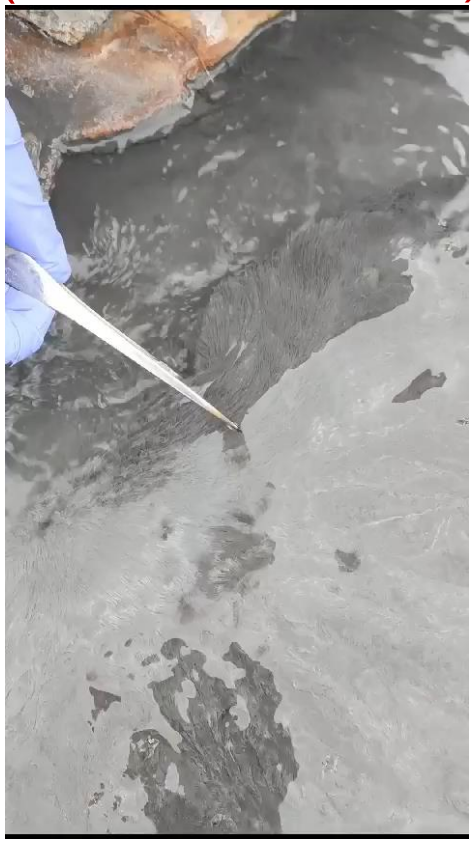
heat protectant

antibiotic / antifungal

surfactant

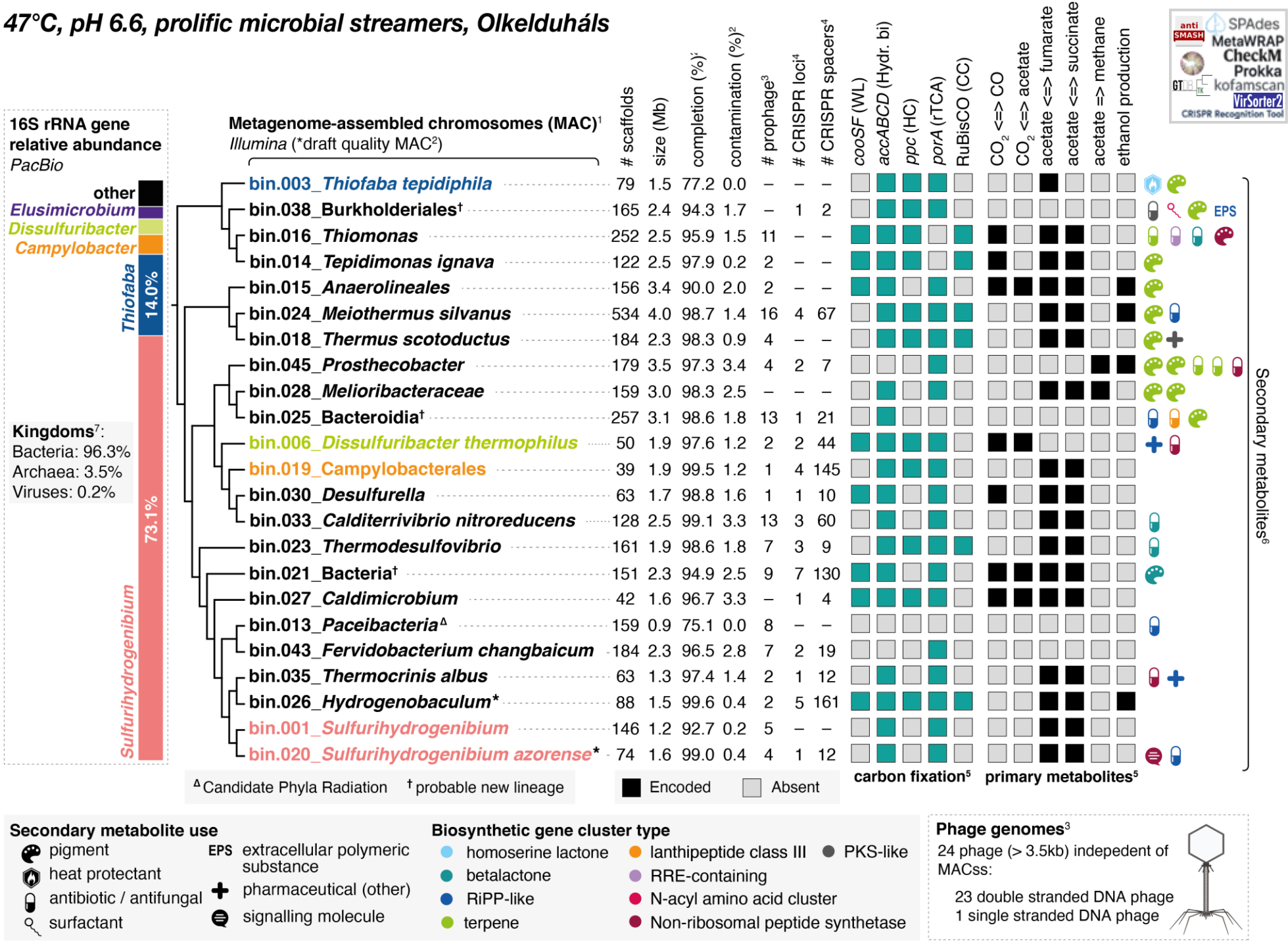


Metagenomics
(*within communities*)



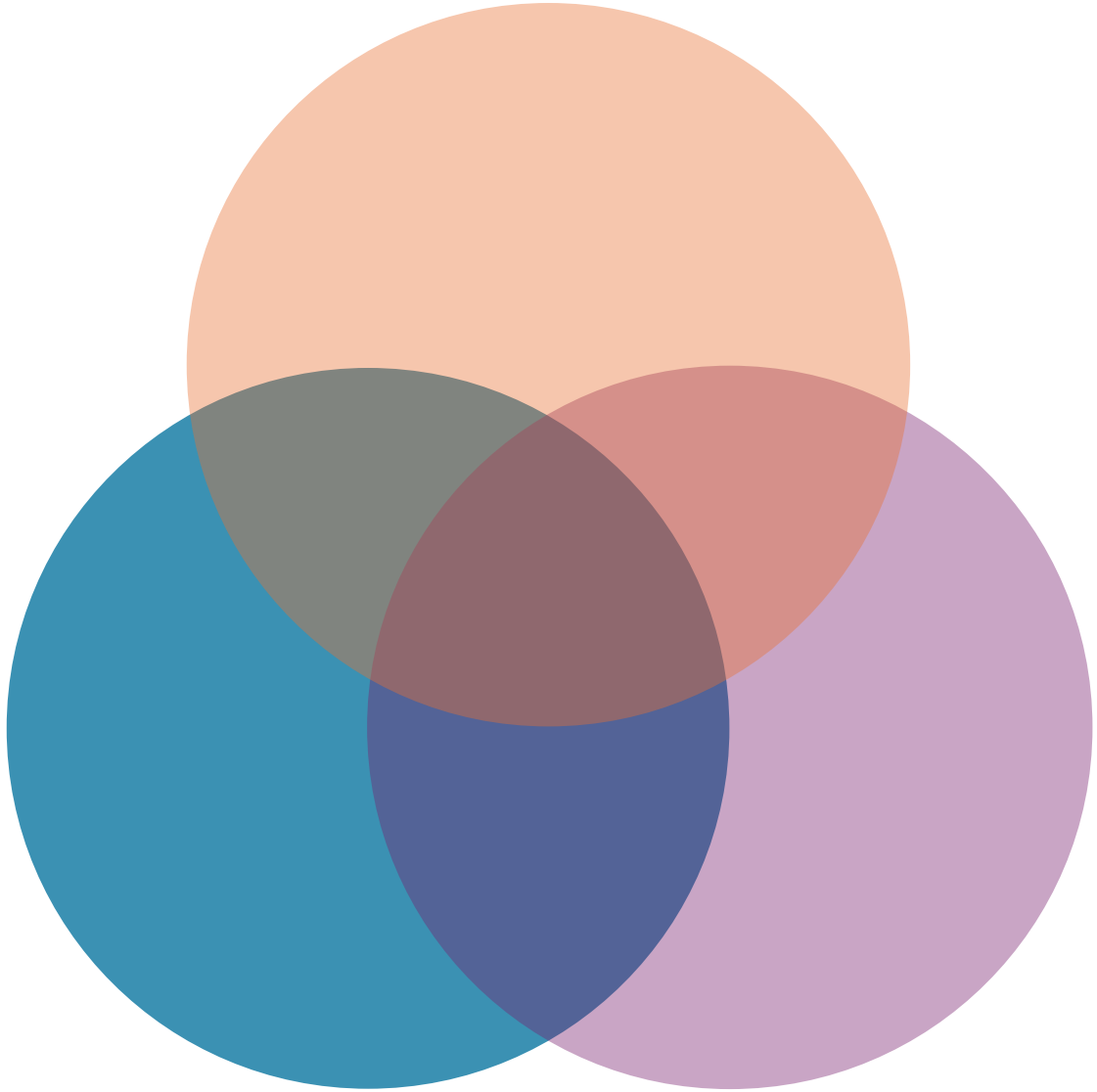
- Strain-level phylogeny
- Draft quality 'genomes'
- New lineages
- Dark Matter
- Metabolic potential
- Viral predation history

47°C, pH 6.6, prolific microbial streamers, Olkelduháls



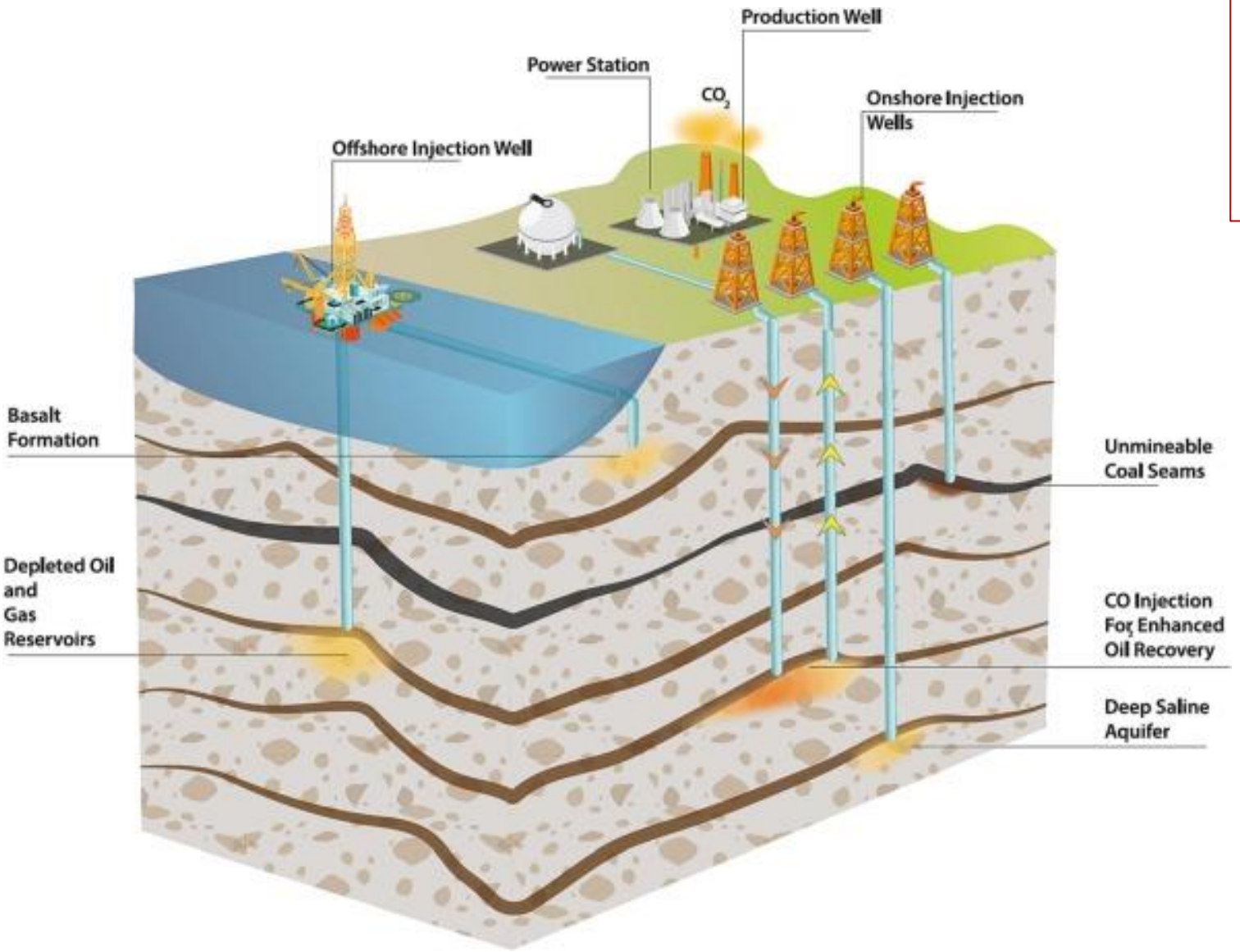
SHORT BREAK

CO_2 Storage



CO_2 ~~Storage~~ (CCUS)

'***storage*** is a misnomer – think of if as ***disposal***"!



ELSEVIER

Earth-Science Reviews

Volume 249, February 2024, 104672

Comprehensive review of CO_2 geological storage: Exploring principles, mechanisms, and prospects

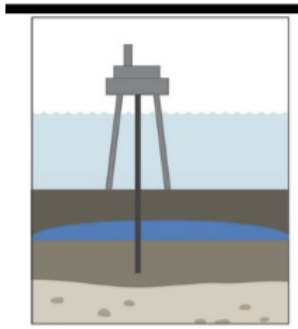
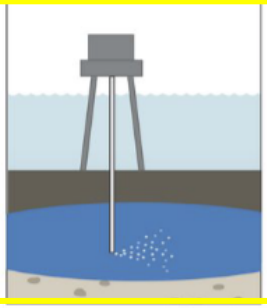
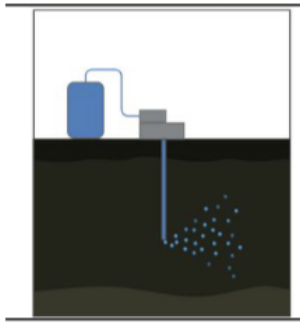
Ahmed Bashir [a], Muhammad Ali [b], Shirish Patil [a], Murtada Saleh Aljawad [a], Mohamed Mahmoud [a], Dhafer Al-Shehri [a], Hussein Hoteit [b], Muhammad Shahzad Kamal [c]

Key attributes for secure geological storage

- **Reservoir geology**
  - porosity
  - permeability
- **Global storage potential**
- **Depth >800m**
- **Containment**
  - caprock integrity, composition
  - trapping mechanisms
- **Accessibility / infrastructure**

Key attributes of the main target formations

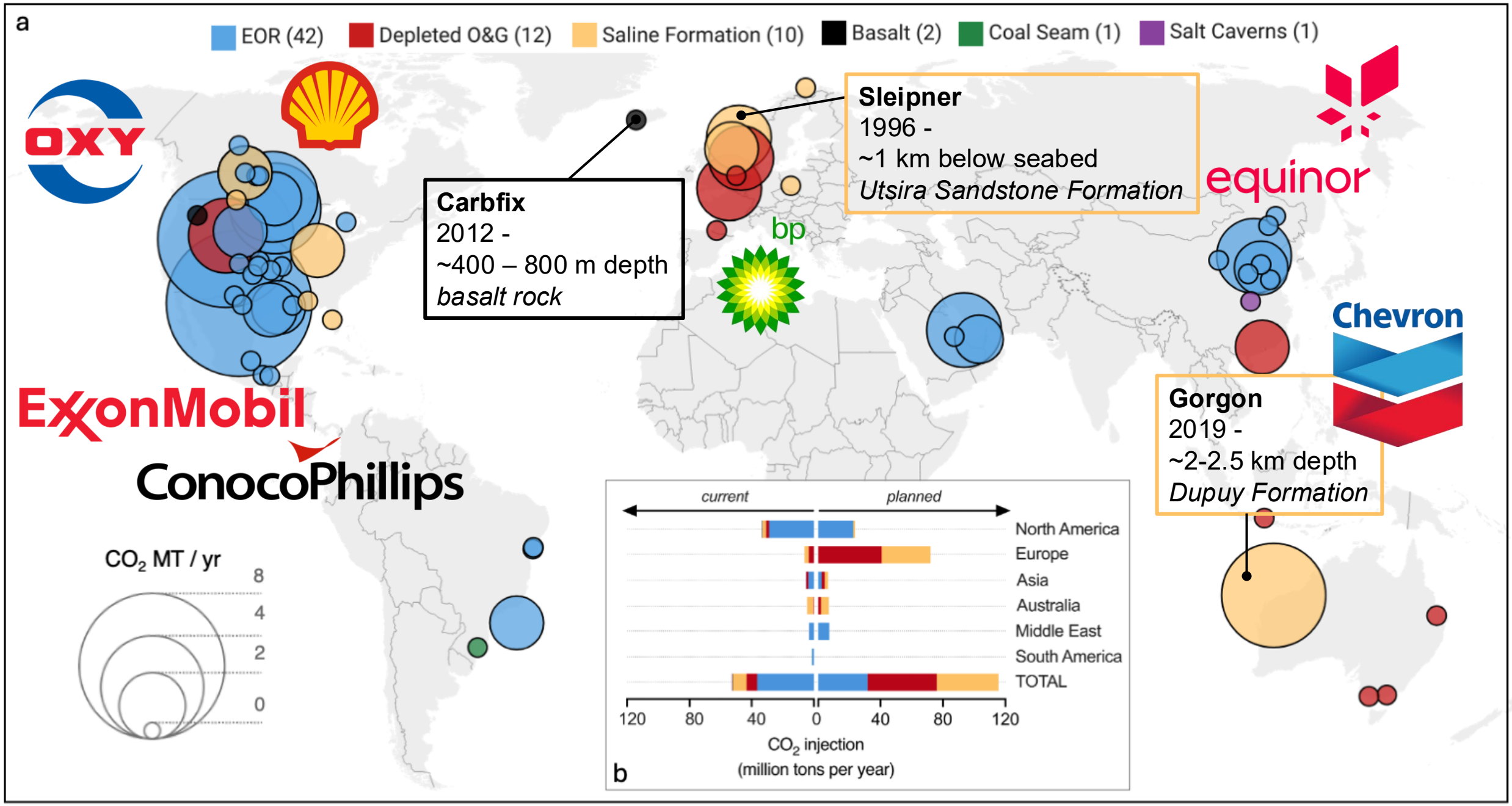
Nixon, Walker and Tyne, *in prep*

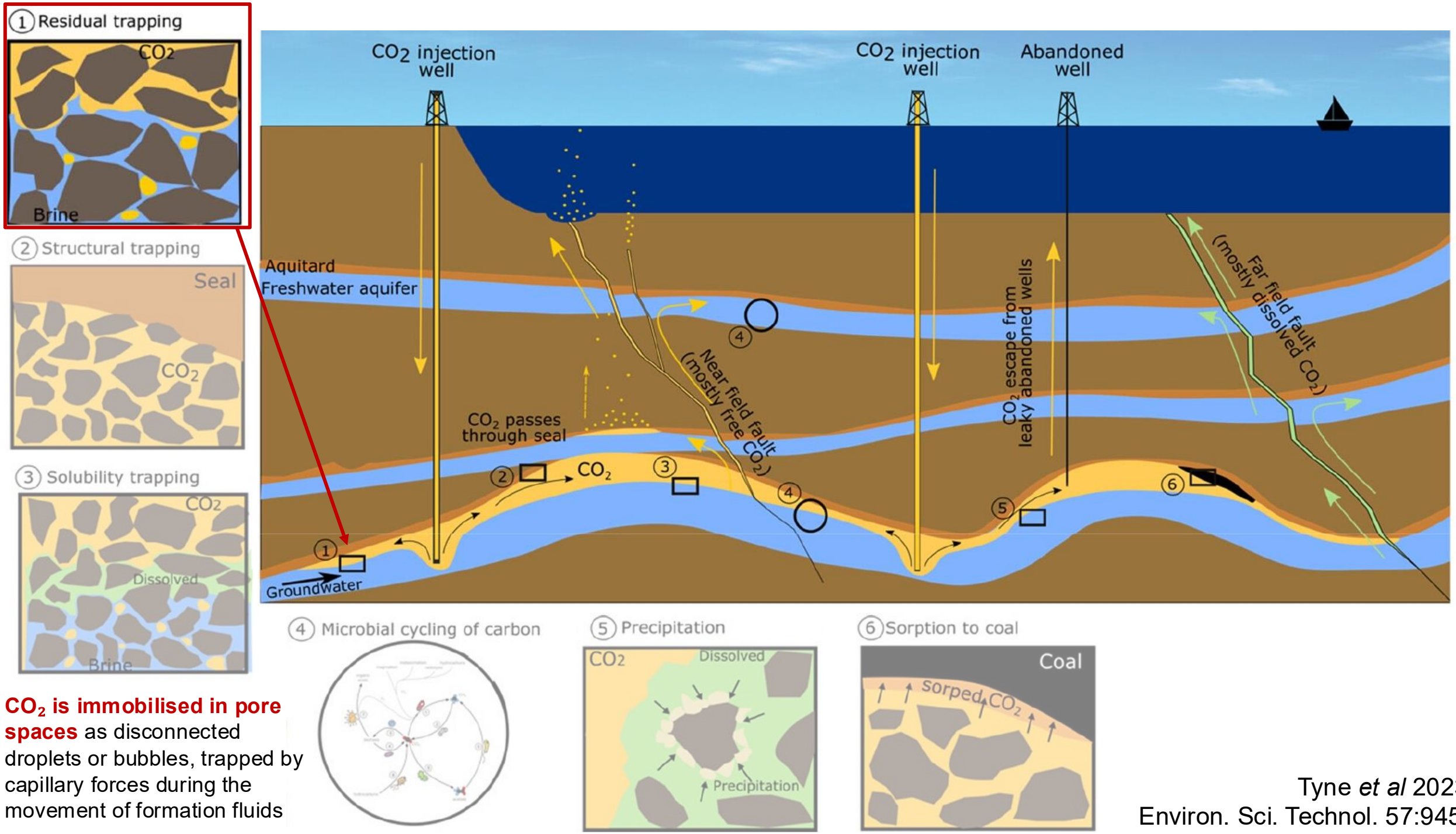
CO_2 Storage Formation Type		Temperature (°C)	Pressure (MPa)	Depth (km)	Salinity (ppm)	Porosity (%)	Permeability (mD)
	**Depleted O&G Reservoir**	30 – 196	<10 – >207	0.1 – 13	10,000 – 200,000	2 – 30	0.02 – >500
	**Saline Aquifer**	20 – 110	19.55 – 42.0	1 - >2.5	100,000 – 160,000	6 – 20	0.02 – >500
	**Basalt Formation**	18 – <300	13.8 – >27.6	0.5 – 1.5	<10,000	6 – 28	0.03 – 3.75

Global capacity: saline aquifer > depleted O&G > basalt

Geological CO_2 storage: *A global perspective*

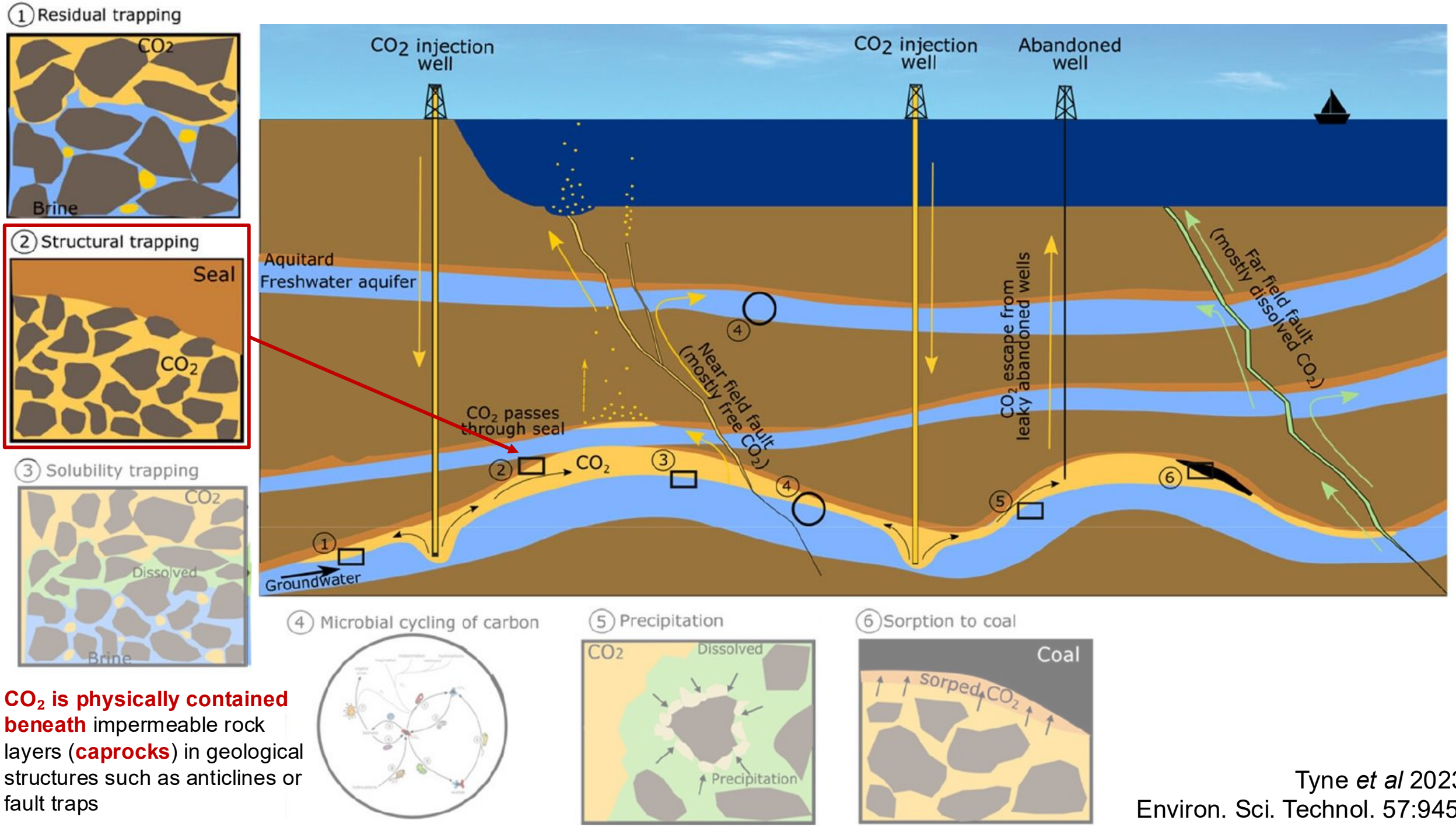
Nixon, Walker and Tyne, *in prep*





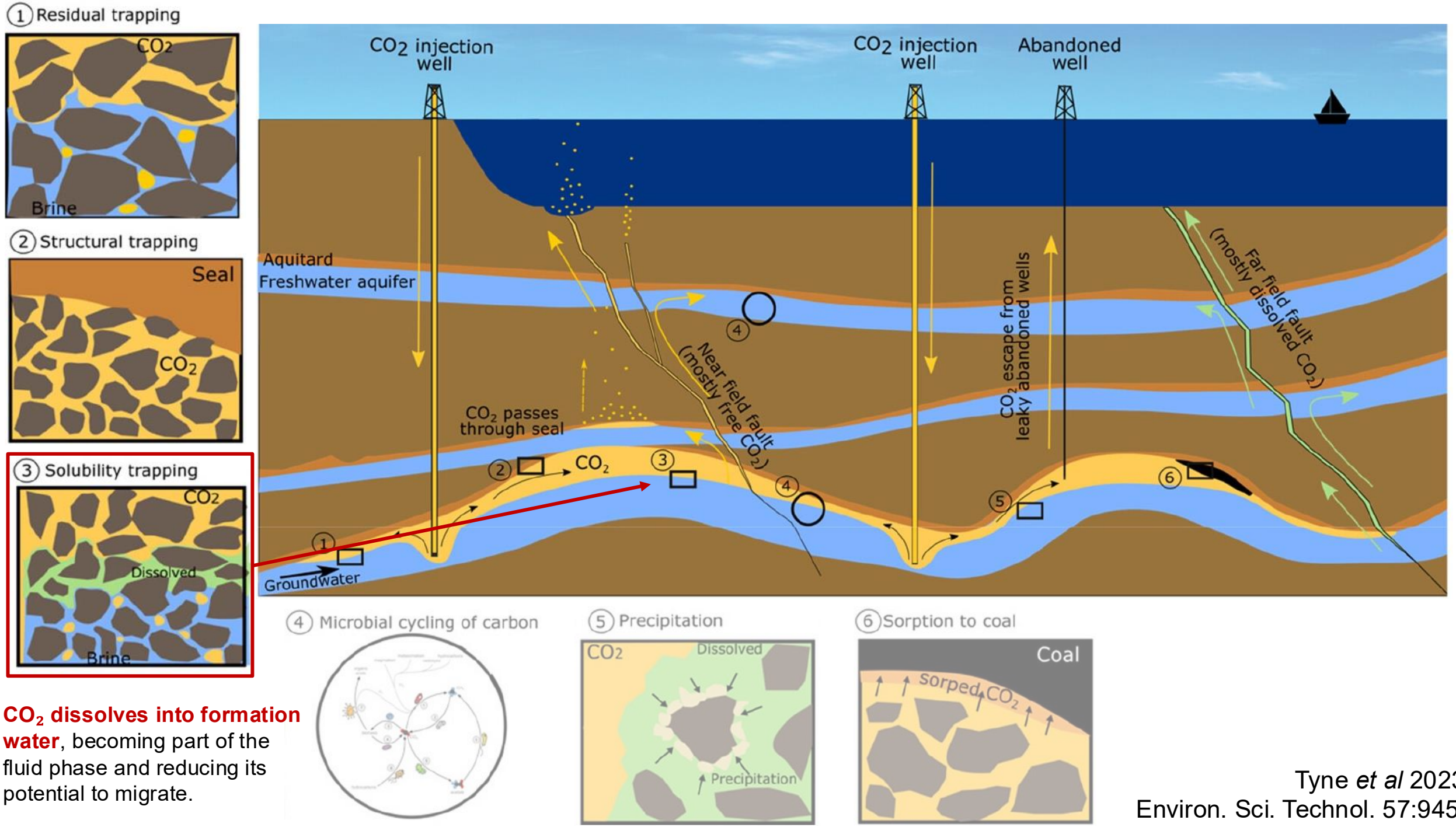
CO_2 is immobilised in pore spaces as disconnected droplets or bubbles, trapped by capillary forces during the movement of formation fluids

Tyne *et al* 2023
Environ. Sci. Technol. 57:9459



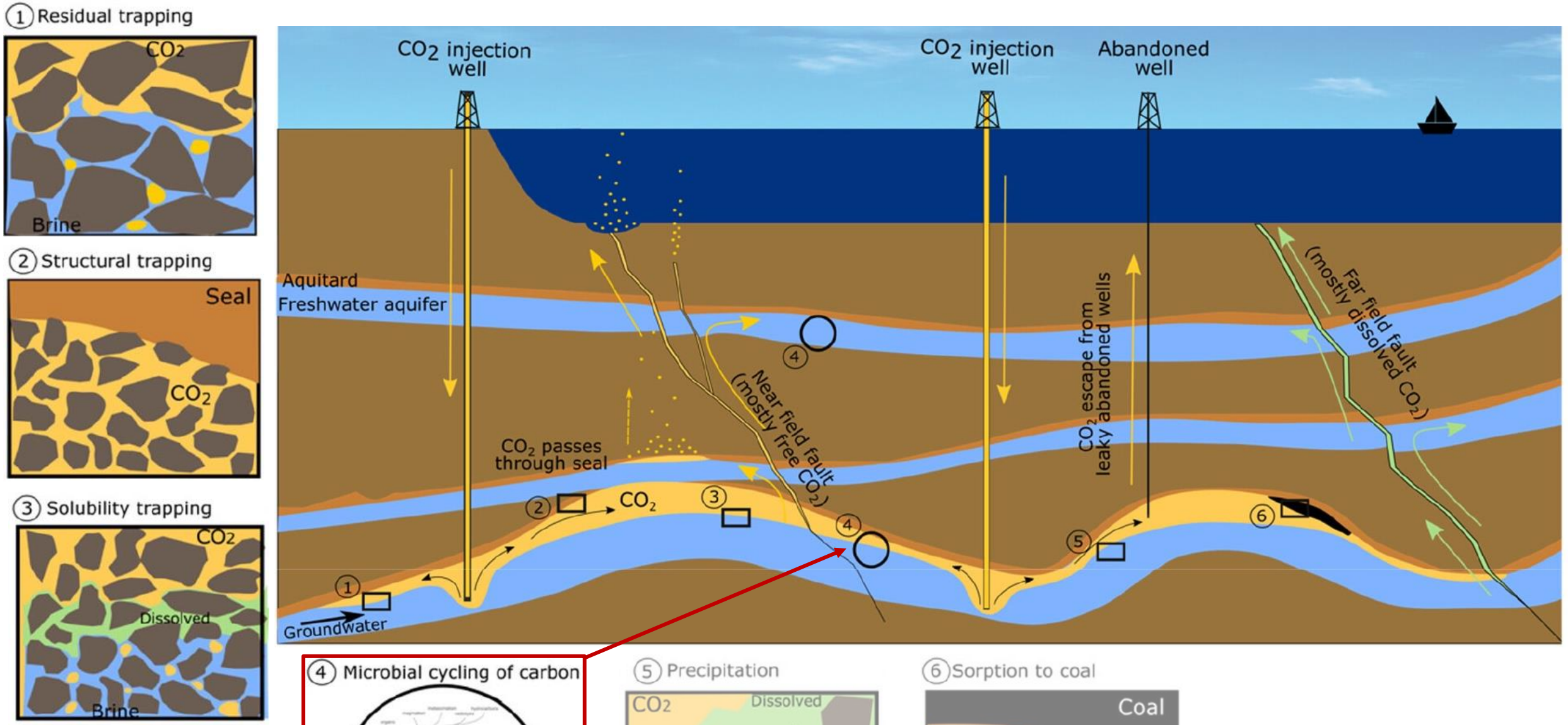
CO_2 is physically contained beneath impermeable rock layers (**caprocks**) in geological structures such as anticlines or fault traps

Tyne *et al* 2023
Environ. Sci. Technol. 57:9459



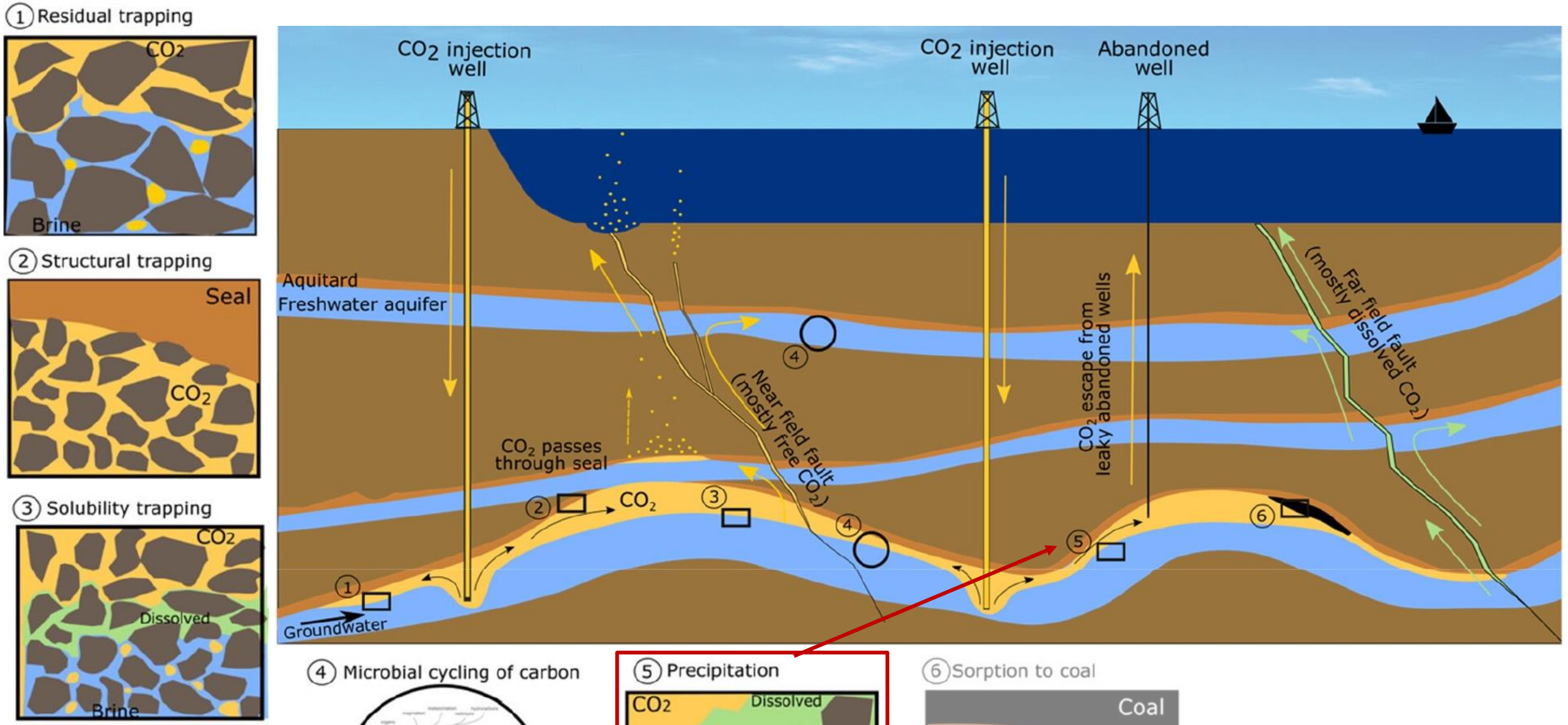
CO_2 dissolves into formation water, becoming part of the fluid phase and reducing its potential to migrate.

Tyne *et al* 2023
Environ. Sci. Technol. 57:9459



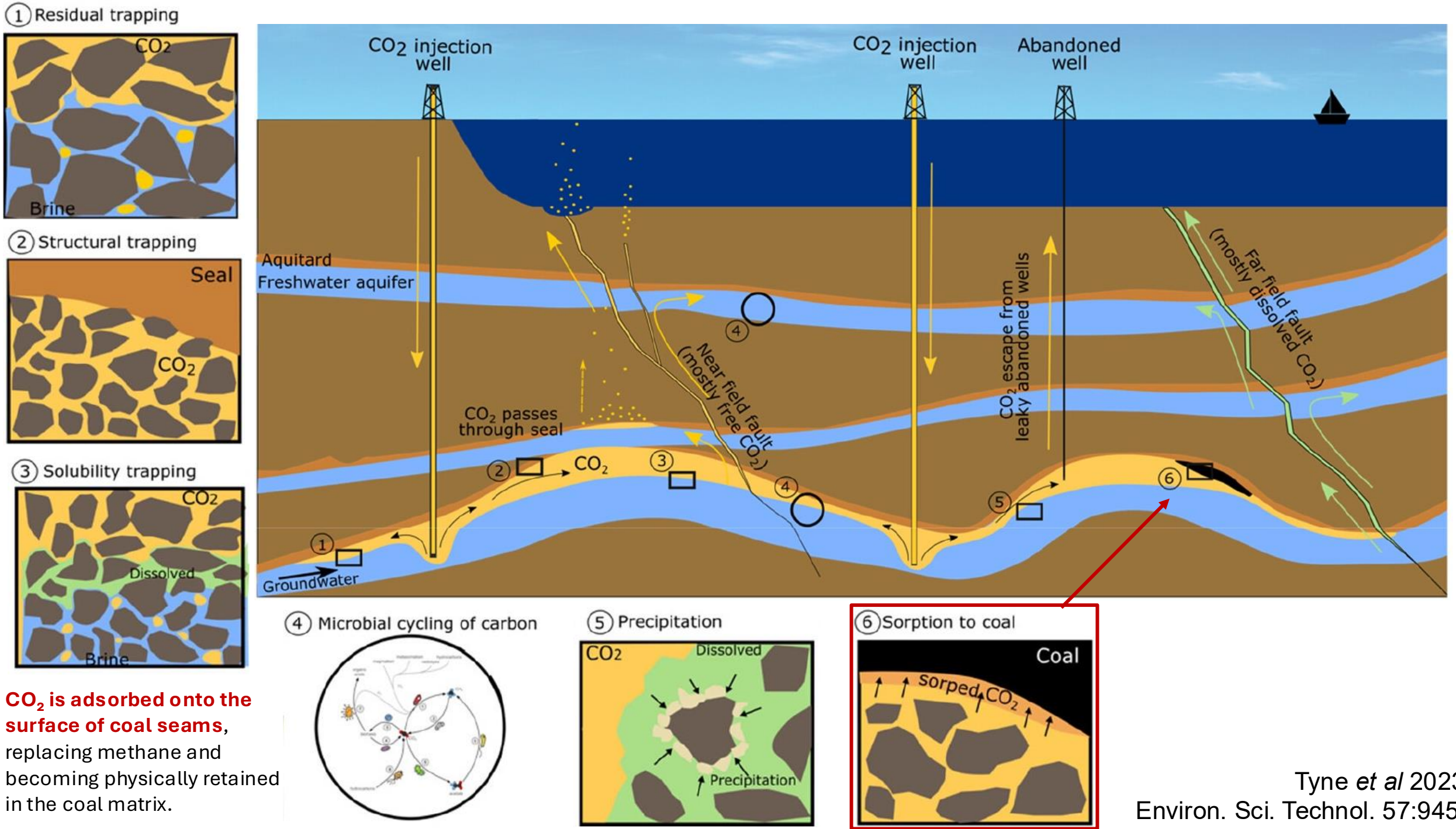
CO_2 is converted into biomass or other forms **through microbial metabolic processes**

Tyne *et al* 2023
Environ. Sci. Technol. 57:9459



CO_2 reacts with minerals in the host rock to **form stable carbonate minerals**, locking carbon away in solid form.

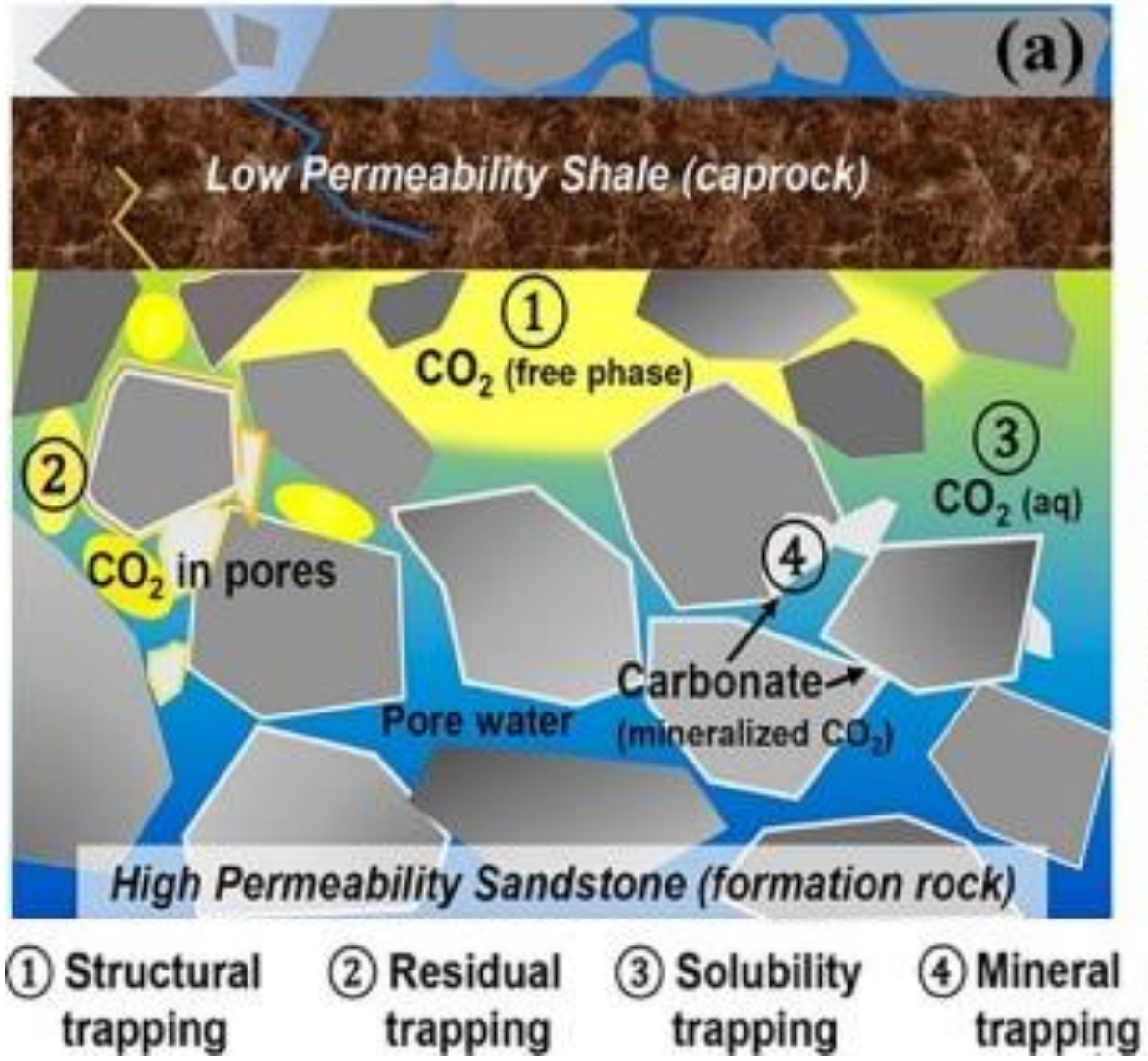
Tyne *et al* 2023
Environ. Sci. Technol. 57:9459

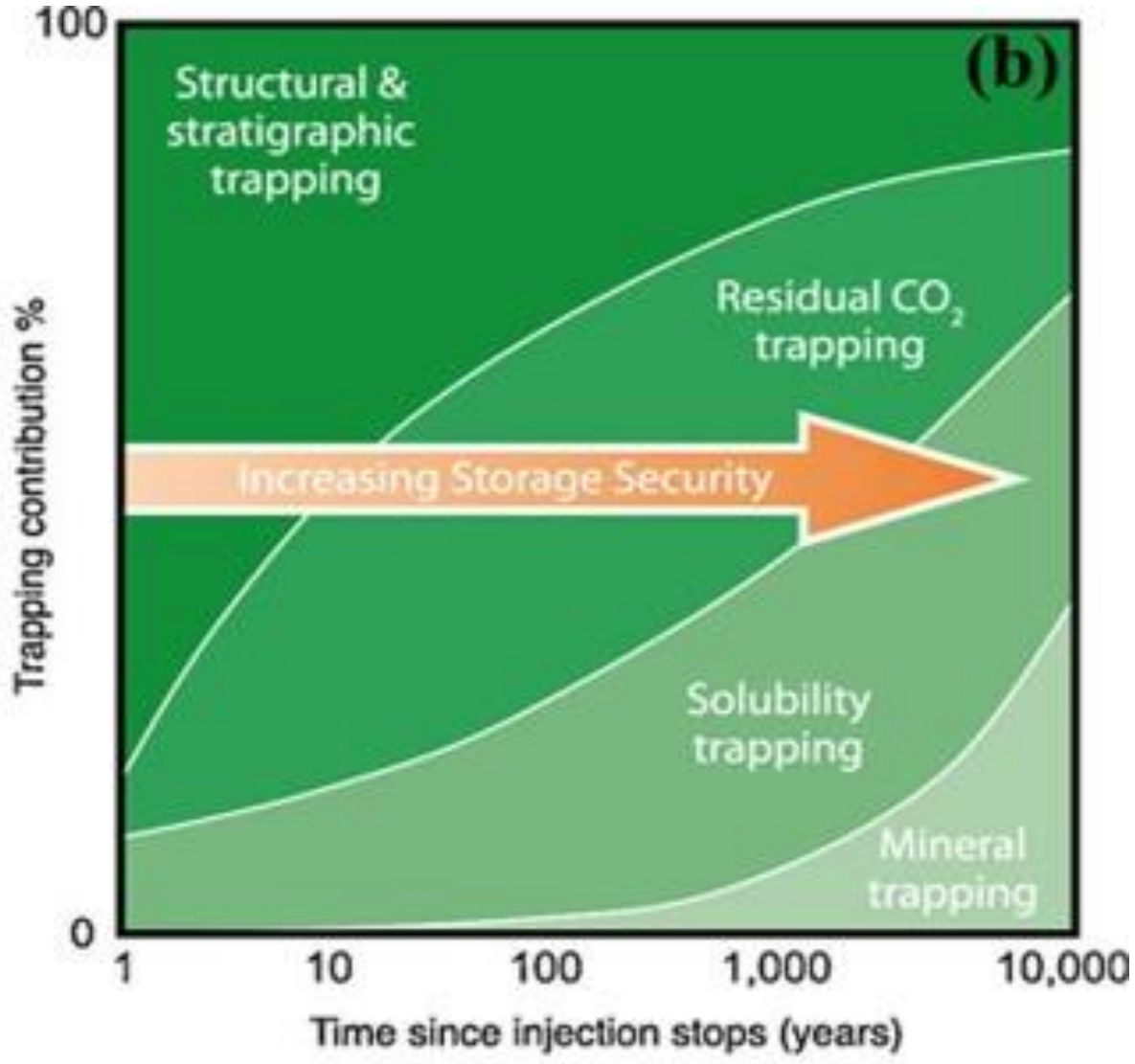


CO_2 is adsorbed onto the surface of coal seams, replacing methane and becoming physically retained in the coal matrix.

Tyne *et al* 2023
Environ. Sci. Technol. 57:9459

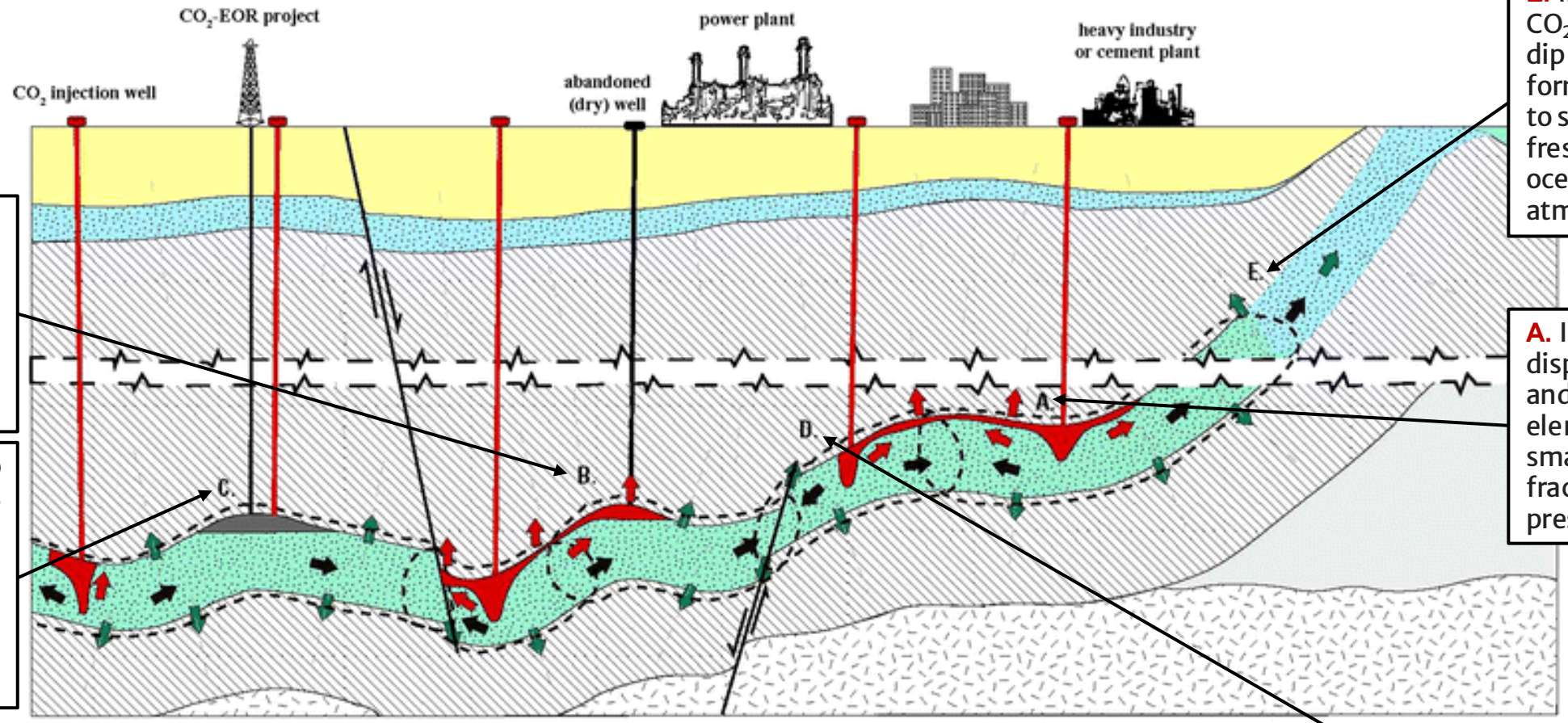
Not all trapping mechanisms are equally secure





Bashir *et al* 2024

Containment risks?



B. Injected CO_2, displaced fluids, and any mobilized elements access faults, pathways to surface, or old abandoned wells

C. CO_2 injected into buoyant traps finds or creates leakage pathways; increases pressure; displaces in situ fluids beyond spill points

E. Injected CO_2 migrates up dip and the storage formation extends to surface bodies of freshwater, oceans, or the atmosphere

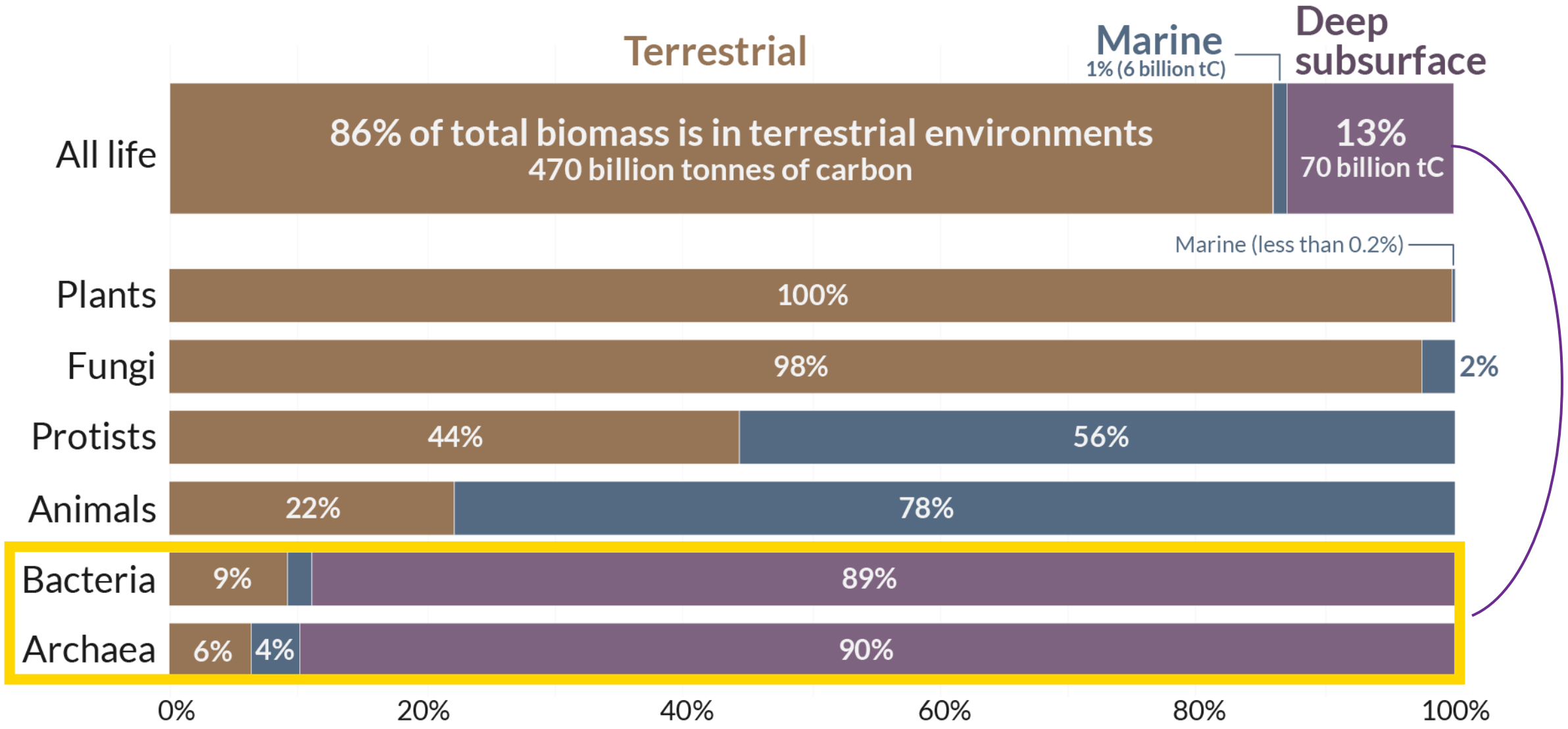
A. Injected CO_2, displaced fluids, and any mobilized elements access smaller faults or fractures; minor pressure buildup.

D. Critical buildup of pressure from injecting CO_2, changes in stress, and a combination of factors that can induce seismicity

carbon dioxide (CO_2)
fresh water
brines
oil or natural gas
unconsolidated sediment
older sedimentary rocks
basement rock
permeable rocks
impermeable rocks
discontinuity in depth
fault
fracture
CO_2 flow
possible formation fluid escape pathways
direction of elevated pressure front(s)

Anderson 2017
Natural Resources Research, **26**:89.

What about the microbiology?



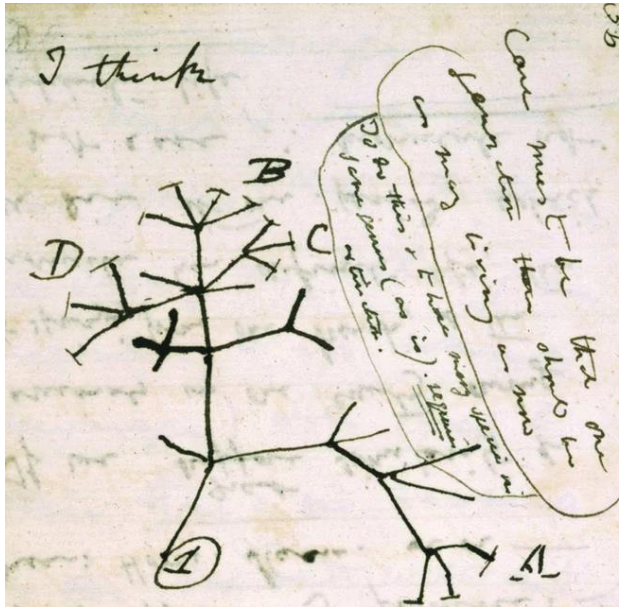
Data source: Bar-On, Y. M., Phillips, R., & Milo, R. (2018). The biomass distribution on Earth. *Proceedings of the National Academy of Sciences.*

OurWorldinData.org – Research and data to make progress against the world's largest problems. Licensed under CC-BY by the authors Hannah Ritchie and Max Roser.

Subsurface life is extraordinarily diverse

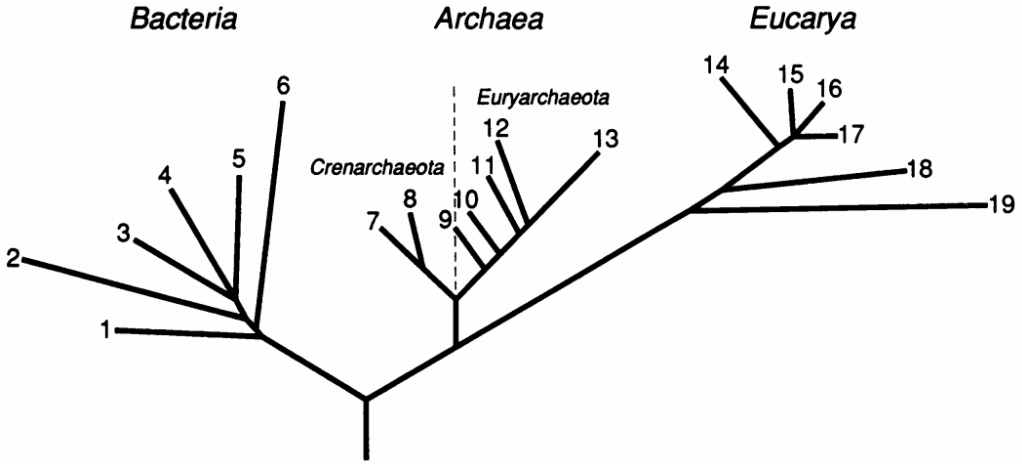
Biological Revolution

Darwin *c.* 1837



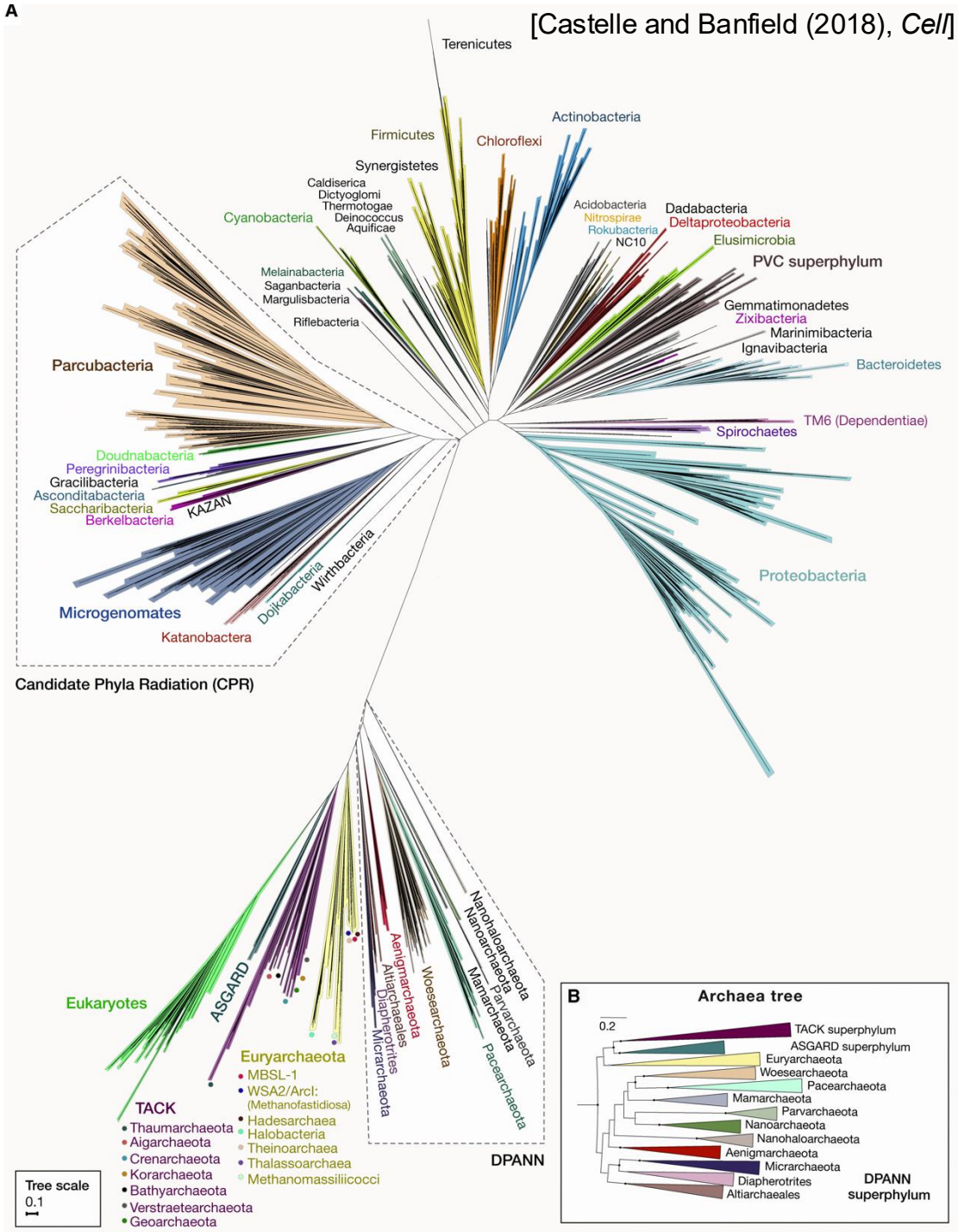
Genetic Revolution

Woese *et al* 1990



Genomic Revolution

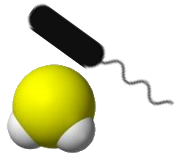
[Castelle and Banfield (2018), *Cell*]



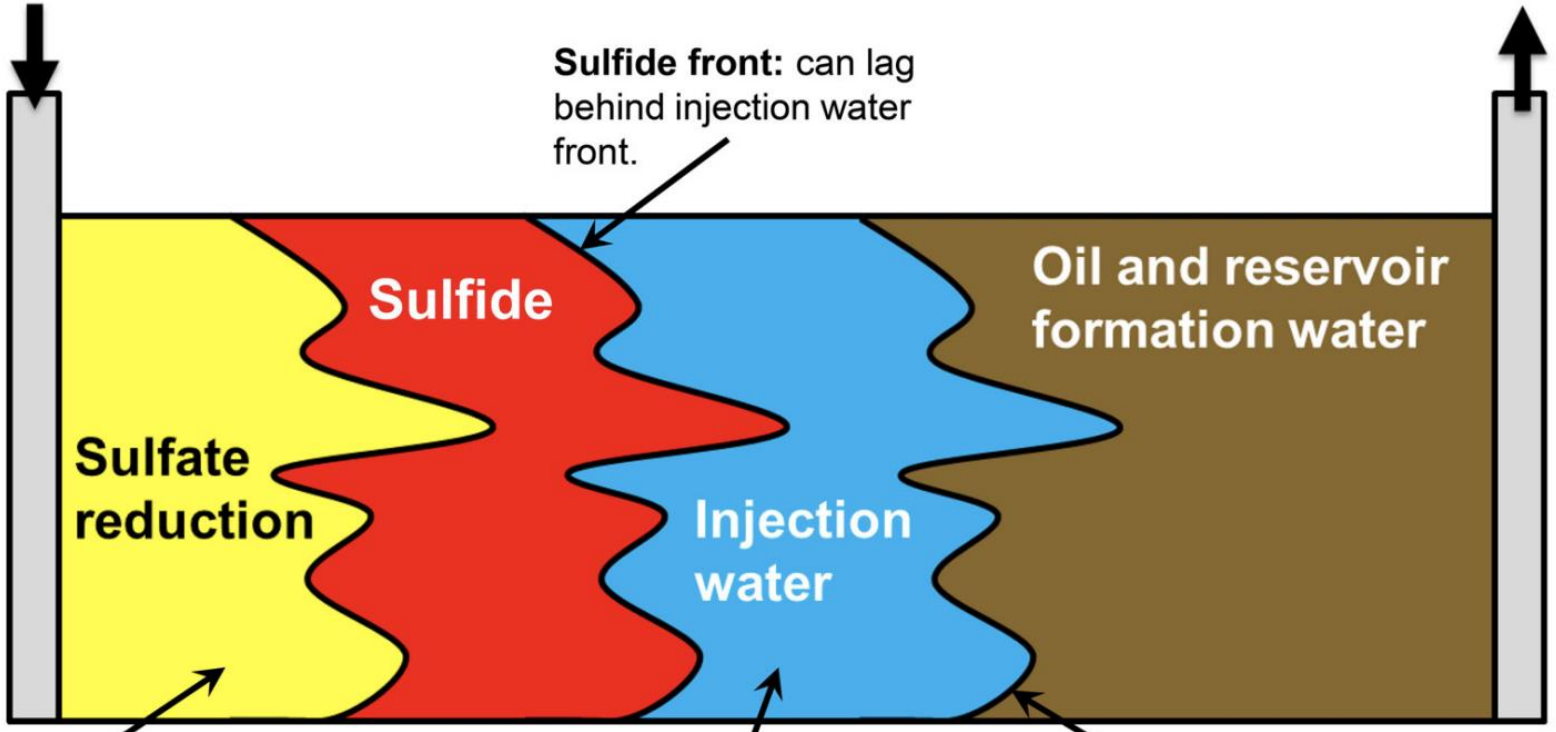
Several subsurface settings are habitats for microbes

Hubbard *et al* 2014

INJECTOR: Water added to maintain reservoir pressure and push-out the remaining oil. Seawater often used - contains abundant sulfate (~28 mM). Produced water can also be re-injected.



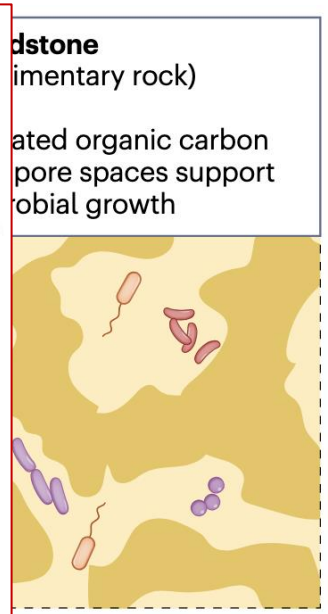
PRODUCER: Oil, water, sulfide.

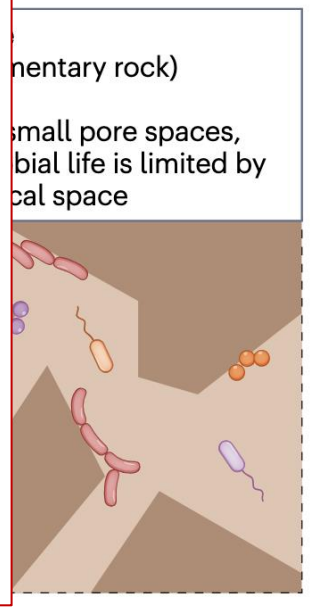


Zone of sulfate reduction: Production of toxic and corrosive sulfide. Horizontal extent likely dependent on reservoir temperature and flow-rate.

Sulfide scavenging zone: Sulfide may be removed from solution by reaction with reservoir minerals (e.g. iron oxides and carbonates) or partitioning into oil phase.

Injection water front: Mixing between injection and reservoir waters can precipitate some sulfate (e.g. as barium sulfate).





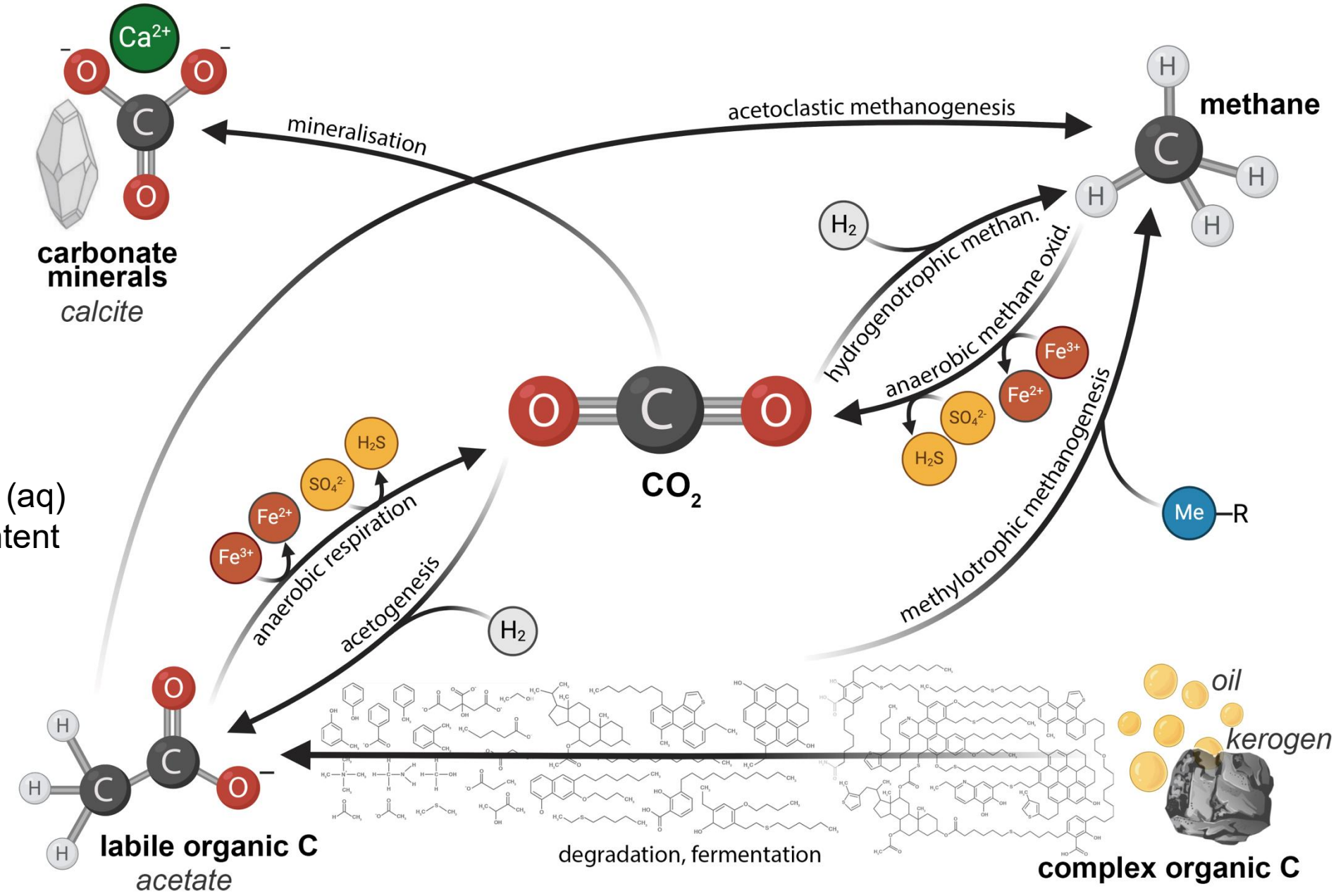
Amundsen *et al* 2024 *Nature Reviews Microbiology*

CO_2 biogeochemistry

Nixon, Walker and Tyne, *in prep*

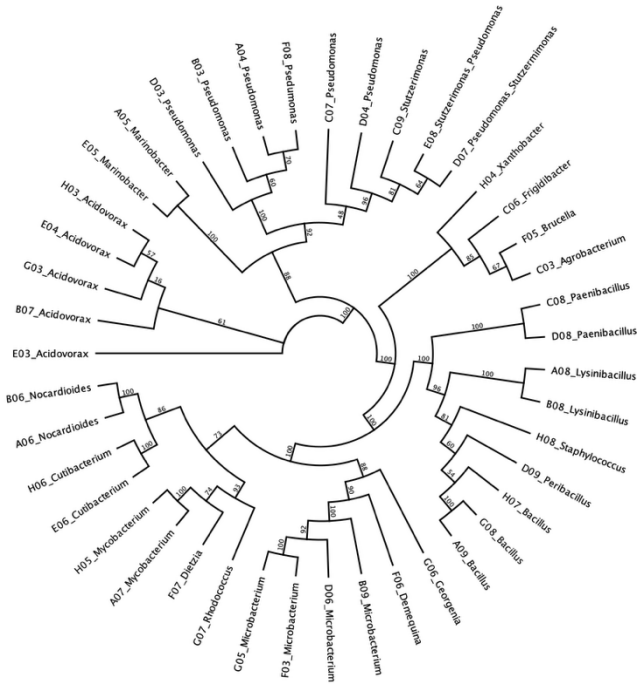
site-specific attributes:

- pH
- redox
- temperature
- pressure
- salinity
- geochemistry (aq)
- organic C content
- **microbiota**

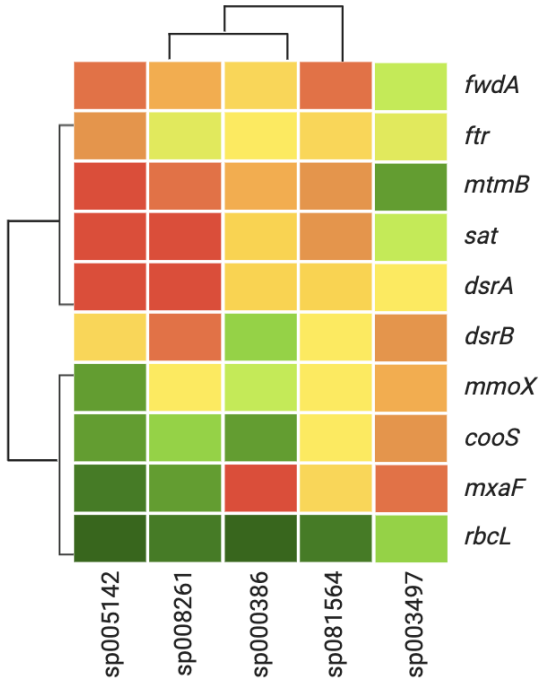


Biogeochemical baselines are essential (and lacking)

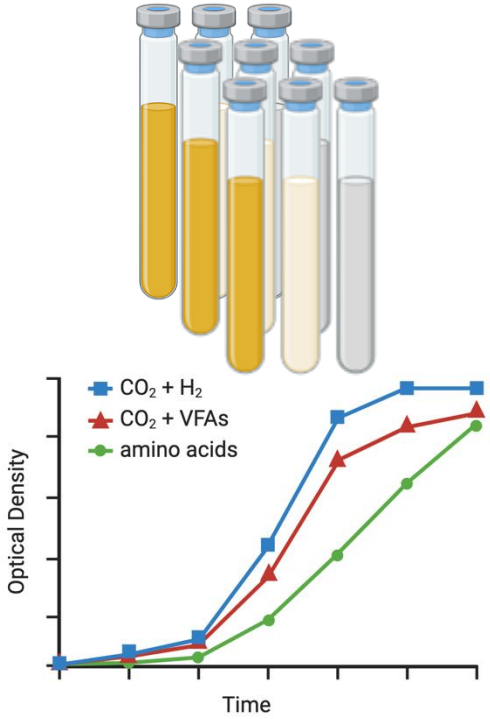
Who?



What?

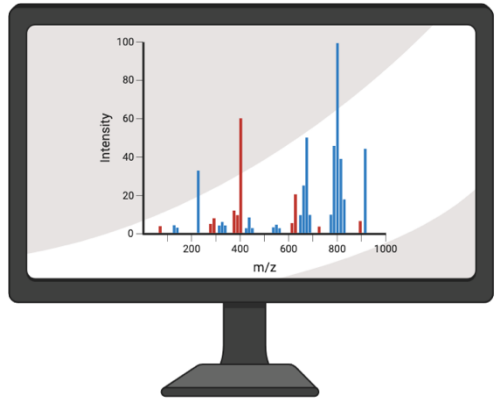


Viable? Active?



in situ conditions?

- pH
- temp
- salinity
- TDS
- pressure
- organic C
- metals
- e^- donors
- e^- acceptors



community membership? biodiversity?

metabolic potential?

metabolic lifestyles? growth rates?

environmental context

CCUS: Summary

What CCUS is and why it matters (ILO 1)

- a suite of tools aimed at the capture, utilisation and permanent storage (disposal!) of CO_2
- CCUS is critical for achieving Net Zero emissions and avoiding the worst climate change impacts

CCUS is an interconnected toolbox (ILO 2)

- Includes a wide range of approaches – nature-based and/or disruptive technologies – designed to reduce CO_2 emissions
- Capture, utilisation and storage are inherently linked – CCUS is not a single solution

Harnessing the natural world to deliver Nature-Based Solutions (ILO 3)

- Established nature-based solutions include:
  - soil carbon sequestration, afforestation / reforestation, biochar, enhanced weathering, habitat restoration
- Emerging disruptive biotechnologies – particularly microbial and microbiome CO_2 sequestration and utilisation – are redefining and evolving nature-based solutions, including at UoM

How CCUS technologies work (ILO 4)

- Capture: direct air capture (DAC), biological fixation, enhanced weathering, biochar…
- Utilisation: CO2 to biofuels and commodity chemicals (e.g. using *Clostridium* and *Halomonas* species, or with pre-evolved microbiomes
- Storage: permanent geological storage targeted at depleted O&G reservoirs, saline aquifers and other (less volumetrically significant) formation types. Safe storage via physical, chemical and biological trapping mechanisms

Uncertainties in CCS (ILO 5)

- Risks: caprock integrity, leakage pathways, deleterious microbial activity (esp. sulfidogenesis, methanogenesis, bioclogging…), monitoring challenges…
- Opportunities: microbially-mediated mineralisation; deep biosphere C-fixation applied to CCU?

Paper of the Week!

nature

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nature > articles > article

Article | Open access | Published: 22 December 2021

Rapid microbial methanogenesis during CO_2 storage in hydrocarbon reservoirs

R. L. Tyne, P. H. Barry, M. Lawson, D. J. Byrne, O. Warr, H. Xie, D. J. Hillegonds, M. Formolo, Z. M. Summers, B. Skinner, J. M. Eiler & C. J. Ballentine

Nature **600**, 670–674 (2021) | Cite this article

42k Accesses | **90** Citations | **180** Altmetric | Metrics



Rebecca Tyne
(now an academic in DEES,

Tyne et al 2021 *Nature*

Glossary of terms

isotope: each of two or more forms of the same element that contain equal numbers of protons but different numbers of neutrons in their nuclei, and hence differ in relative atomic mass but not in chemical properties; in particular, a radioactive form of an element

isotopic fractionation: the separation or enrichment of one isotope of an element over another during a natural process. Methanogens preferential use lighter isotopes of CO_2 (^{12}C) over heavier isotopes (^{13}C) during methane production, resulting in a relative depletion of ^{13}C in biogenic methane (and enrichment in ^{13}C in the residual CO_2 pool)

clumped isotope: heavy isotopes that are bonded to other heavy isotopes. methane clumped isotope can record its formation temperature, and therefore can be used to identify the origins of methane

noble gases / isotopes: The noble gases are the members of group 18 of the periodic table: helium (He), neon (Ne), argon (Ar), krypton (Kr), xenon (Xe), radon (Rn) and, in some cases, oganesson (Og). The relative isotopic abundances of noble gases serve as an important geochemical tracing tool in earth science

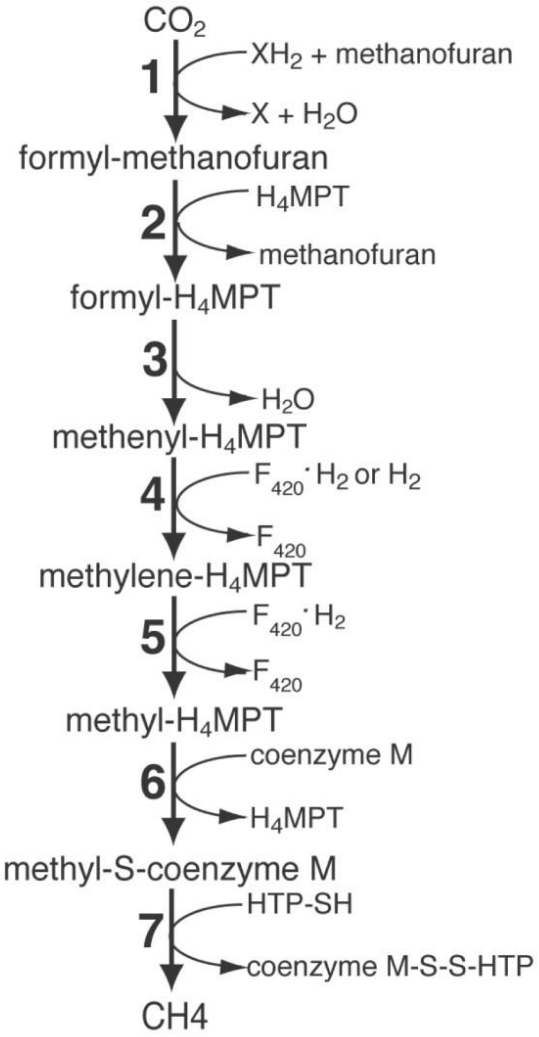
methanogenesis: the formation of methane coupled to energy conservation by Archaea known as methanogens

hydrogenotrophic methanogenesis: microbially-mediated conversion of CO2 and H2 to methane (below and right)

$$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$$

AOM (Anaerobic Oxidation of Methane): microbial methane consumption in the absence of oxygen; part of the broader methane-implicated carbon cycle in the study site

CO_2-EOR (Enhanced Oil Recovery): a technique used to extract additional oil from reservoirs after primary and secondary recovery methods have become ineffective, by injecting CO_2



Tyne et al 2021 *Nature*

Abstract

Carbon capture and storage (CCS) is a key technology to mitigate the environmental impact of carbon dioxide (CO_2) emissions. An understanding of the potential trapping and storage mechanisms is required to provide confidence in safe and secure CO_2 geological sequestration[1,2]. Depleted hydrocarbon reservoirs have substantial CO_2 storage potential[1,3], and numerous hydrocarbon reservoirs have undergone CO_2 injection as a means of enhanced oil recovery (CO_2-EOR), providing an opportunity to evaluate the (bio)geochemical behaviour of injected carbon. Here we present noble gas, stable isotope, clumped isotope and gene-sequencing analyses from a CO_2-EOR project in the Olla Field (Louisiana, USA). We show that microbial methanogenesis converted as much as 13–19% of the injected CO_2 to methane (CH_4) and up to an additional 74% of CO_2 was dissolved in the groundwater. We calculate an in situ microbial methanogenesis rate from within a natural system of 73–109 millimoles of CH_4 per cubic metre (standard temperature and pressure) per year for the Olla Field. Similar geochemical trends in both injected and natural CO_2 fields suggest that microbial methanogenesis may be an important subsurface sink of CO_2 globally. For CO_2 sequestration sites within the environmental window for microbial methanogenesis, conversion to CH_4 should be considered in site selection.

Tyne et al 2021 *Nature*

The **Olla Oil Field** (Louisiana, in the northern Gulf of Mexico) was CO_2 injected between 1983 and 1986 for enhanced oil recovery (CO_2-EOR)

- total 2.2×10^8 m^3 of CO_2 was injected; ~9×10^7 m^3 remained in the reservoir after EOR project ended
- water flooding in 1984

This study integrates noble gas and clumped isotope data with stable isotope and microbiological data to investigate the fate of the injected CO_2

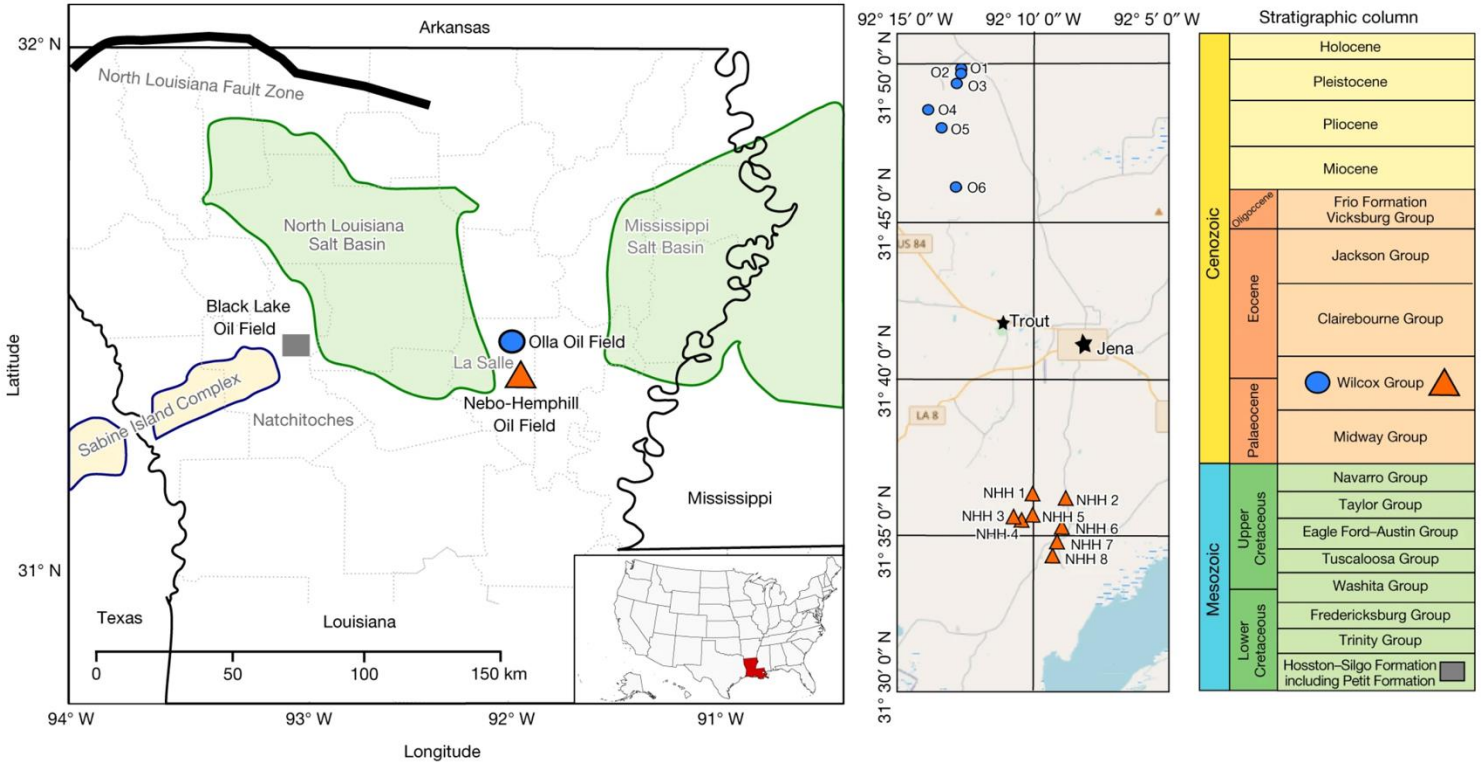
The Nebo-Hemphill Oil Field (20 km SE of Olla) served as a control, having had no CO_2 injected

- significantly higher CO_2 concentrations in the injected Olla Oil Field (18 ± 12mol%) vs the Nebo-Hemphill Oil Field (1.02 ± 0.69 mol%)
- CH_4 concentrations at Olla are 77 ± 13 mol% (STP) compared with 92.6 ± 8.2 mol% (STP) at Nebo-Hemphill

Stable isotopes of carbon are frequently modified by chemical or biological processes; the inert nature of noble gases means that they are ideal tracers for injected CO_2

Fig. 1: Map of study area showing locations of the Olla and Nebo-Hemphill oil fields as well as the Black Lake Oil Field, from which the injected CO_2 was sourced.

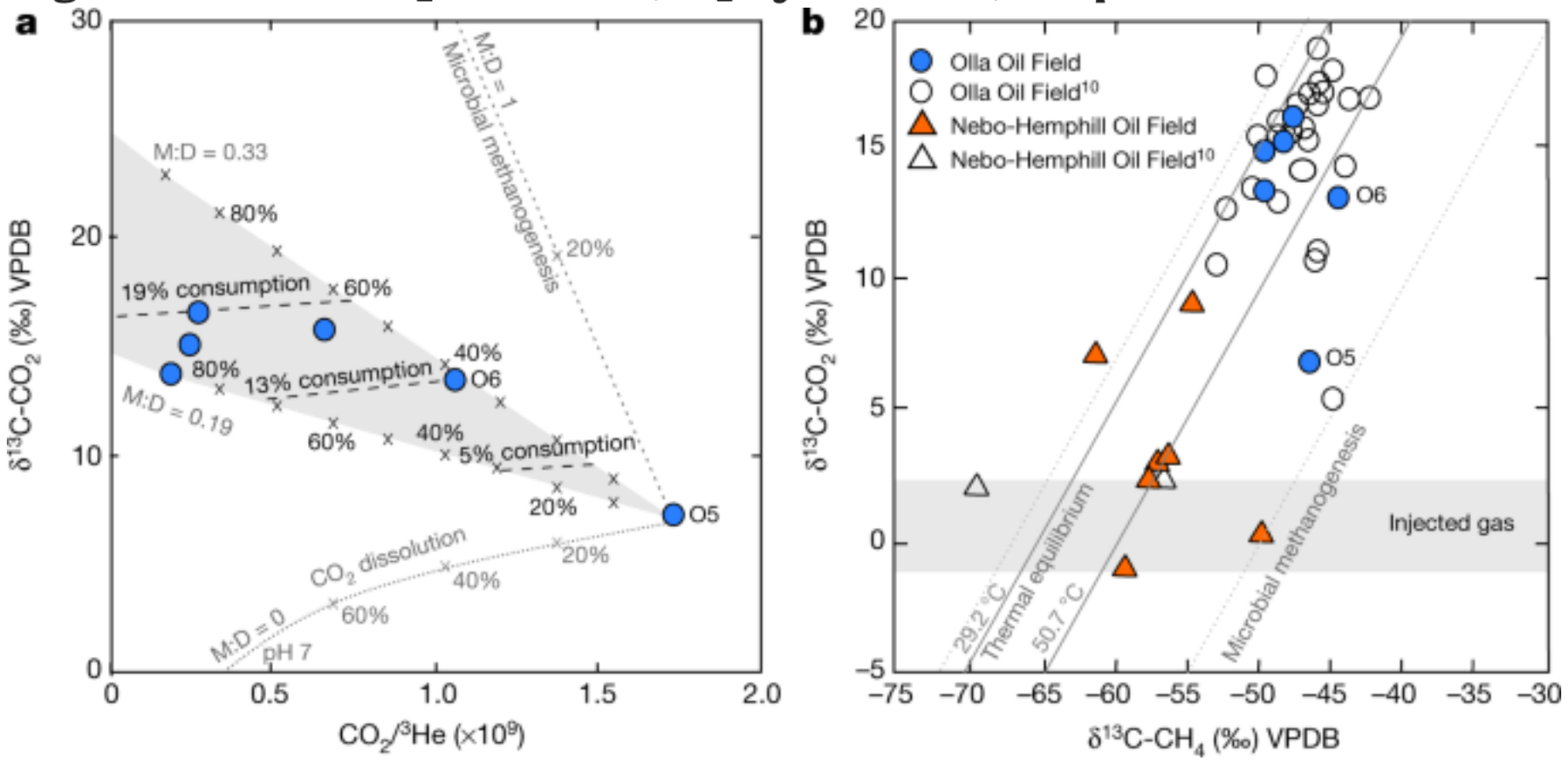
From: Rapid microbial methanogenesis during CO_2 storage in hydrocarbon reservoirs



Only the Olla Oil Field contains injected CO_2. The inset on the right shows an expanded view with individual sample locations (nearby urban areas are denoted by stars) and a stratigraphic column showing the relevant lithologic units[9].

Tyne et al 2021: $\delta^{13}C$ of CO_2

Fig. 2: The $\delta^{13}C$ of CO_2 in the Olla (CO_2 injected field) samples.



a $\delta^{13}C$ of CO_2 as a function of the $CO_2/^3He$ ratio. The dashed lines show endmember methanogenesis and dissolution (pH 7) fractionation trajectories. The shaded region represents trapping by the combination of both microbial methanogenesis and dissolution. **Dissolution accounts for approximately three times more CO_2 removal (M:D = 0.26) than microbial methanogenesis**. The lines labelled 'consumption' show the portion of original injected CO_2 that has been removed by net microbial methanogenesis.

b $\delta^{13}C$ of CO_2 as a function of the $\delta^{13}C$ of CH_4. The shaded region represents the CO_2 isotopic composition of the injectate from the Black Lake Oil Field into the Olla system. The Olla data are consistent with thermal re-equilibration with both reservoir temperatures (solid lines) and microbial methanogenesis (dotted lines). The 1σ level of uncertainty is within the symbol size.

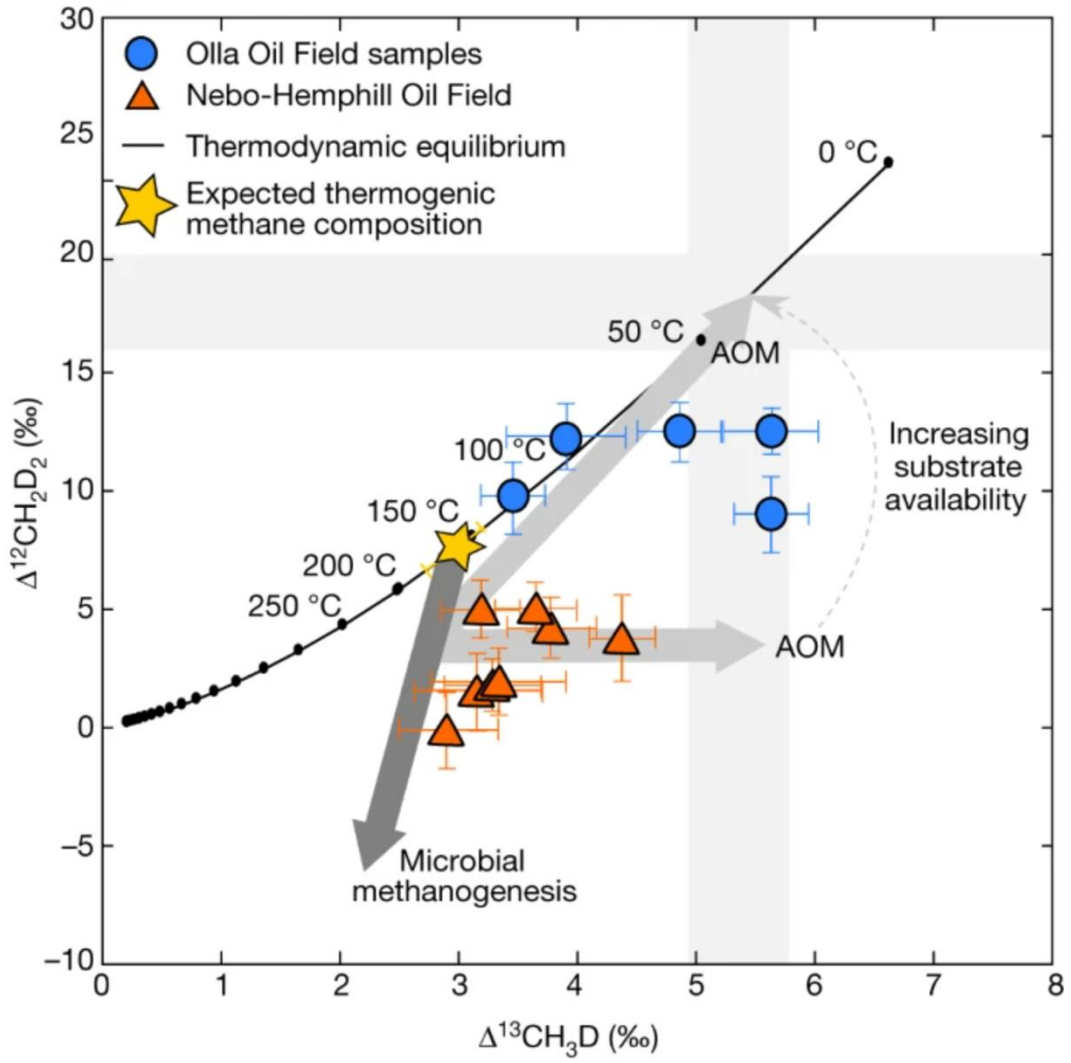
Tyne et al 2021: clumped isotopes

Clumped isotope compositions provide an **independent constraint on the origin of CH_4 in petroleum systems**. Thermogenic CH_4 appears to dominantly form under internal isotopic equilibrium whereas microbial CH_4 is highly variable in its clumped isotope compositions, with both non-equilibrium and apparent equilibrium signatures

→ Clumped isotopologues provide clear evidence that the Olla **CH_4 is microbial in origin**

We conclude that although methanogenesis is not the dominant CO_2 sink, it can represent a substantial process within both natural and perturbed CO_2 fields and may be significantly more prevalent than previously considered. Even with the most conservative estimates of CO_2 conversion by methanogenesis in the Olla Oil Field, we find that, so far, as much as 13% to 19% of the emplaced CO_2 in sampled sections of the field has been consumed by methanogens. This is less than the amount of CO_2 that has been dissolved into water, but similarly occurs at significant rates on engineering timescales. CH_4 is less soluble and more mobile than CO_2 and therefore there is an enhanced risk of gas loss associated with CH_4 production due to microbial methanogenesis. Depleted hydrocarbon reservoirs have the second-largest CO_2 storage potential[1,3] after deep saline aquifers. If physicochemical and environmental conditions within these reservoirs (for example, low temperature, low salinity and high substrate availability) are conducive to microbial methanogenesis, we suggest that these microbial processes should be considered as criteria for future CCS site selection to ensure low-risk long-term storage. In addition, integrated studies like this may prove essential to effectively monitor CO_2 storage and the biogeochemical processes that result as a consequence of it.

Fig. 3: $\Delta^{12}CH_2D_2$ versus $\Delta^{13}CH_3D$ of the measured Olla and Nebo-Hemphill samples.

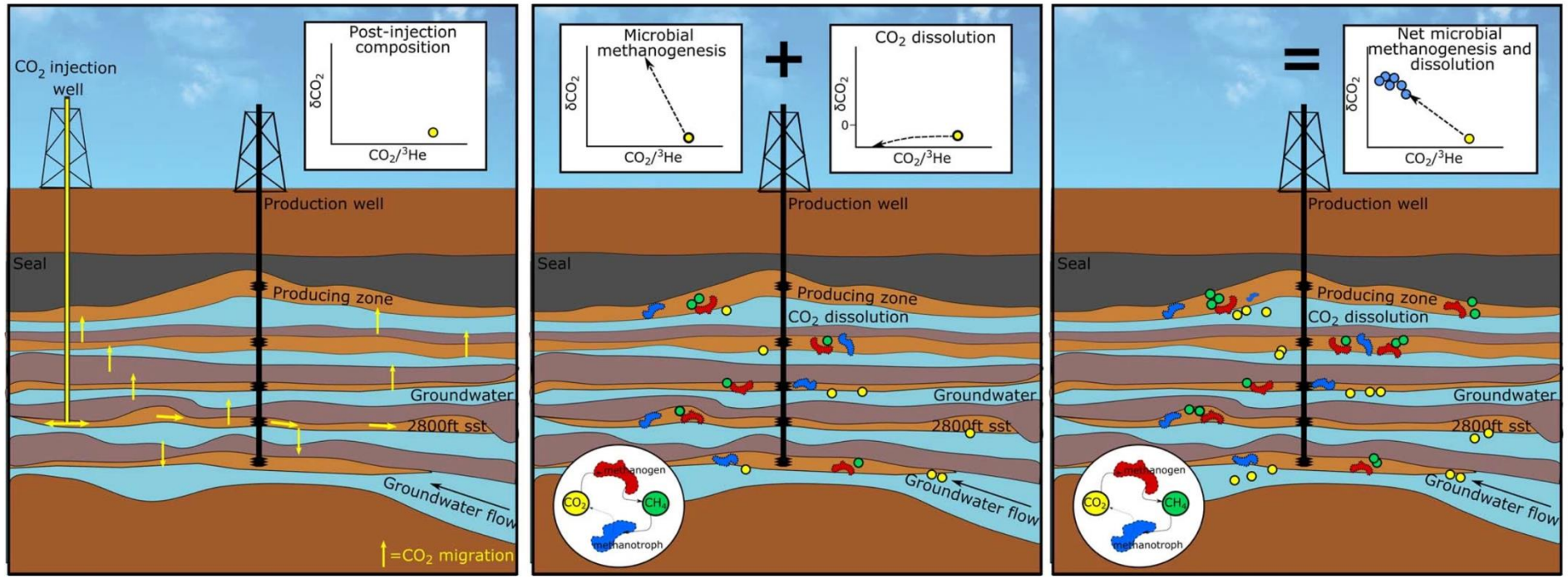


The clumped isotopologue space illustrates whether measured CH_4 is at internal thermodynamic equilibrium (black line) or not. The shaded cross represents thermal equilibration to current reservoir temperatures in each isotopologue. The arrows represent the theoretical trends for methanogenesis (dark grey) and AOM (light grey)

Tyne et al 2021: authors' interpretation

Extended Data Fig. 4: Schematic of the processes occurring within the Olla Oil Field and resulting changes in the isotopic composition within the field.

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The first panel shows CO_2 injection into the 2,800-ft sandstone (sst) and subsequent lateral and vertical migration (yellow arrows) to all the producing formations within the field. The middle panel demonstrate the different processes (microbial methangonesis, AOM and dissolution), which are occurring in the field and their effects on the isotopic composition. Yellow circles represent CO_2 dissolution and green circles are the net CH_4 production from microbial activity. Hydrogen for microbial methanogenesis is sourced from the hydrocarbons and water. The final panel shows the current state of the reservoir.

Discuss:

1. Is the microbial conversion of methane in a geological CO_2 storage reservoir a risk or an opportunity? Why?

2. Is residual oil important in this scenario? How might this finding relate to CO_2 stored in saline aquifers?

3. What could be done in future CO_2 injection campaigns to anticipate this scenario?